



Practical Exercises in Research Methods

Macro Drivers

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report date:

10/05/2026

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I. Abstract

This paper examines the extent to which housing prices in 16 European countries deviate from a value that can be explained by macroeconomic fundamentals. The effective VECM sample comprises 15 countries, the methodological reason for the exclusion of Denmark is given in §4.5. It also examines whether the estimated model can forecast price dynamics on a purely macroeconomic basis. Investment decisions are explicitly outside the scope of this paper.

The empirical approach follows a country specific time series logic. For each country, a system with up to four dimensions is built. It contains logged real house prices and, where the ADF based $I(1)$ filter confirms that they are non-stationary, logged GDP per capita, inflation and the 10-year government bond yield.

Methodologically, the paper combines unit root tests, Johansen cointegration tests and vector error correction models. It therefore explicitly separates long run equilibrium relationships from short run dynamics. This separation is crucial because property prices are highly persistent and a pure level-based view can easily create a false impression of precision. Model quality is assessed through stability and residual autocorrelation. Countries with stable dynamics and unobtrusive LM diagnostics are classified as Solid. Countries with stable but autocorrelated residual dynamics are classified as Weak.

The real house price specification is used consistently for all countries in which a real cointegration relationship can be identified. For Hungary and Sweden, where the Johansen test in the real specification did not deliver a rank, we report the nominal estimate as a fallback and mark these countries accordingly in the results tables. The forecast evaluation does not include a formal benchmark comparison against atheoretical time series models. This boundary follows from the macroeconomic identification scope of the paper.

1 Introduction

1.1 Research Aim

The European residential property market is not only an asset market. It also acts as a macroeconomic transmission system. Price movements influence household wealth, collateral values, construction activity and risk perception. In this way, they can affect the business cycle and financial stability (Cuestas et al., 2023, p. 3216 ff.; Girouard et al., 2006, p. 6-7, p. 28 ff).

The aim of this paper is to test, for 16 European countries, whether residential property prices systematically deviate from an empirically estimated fundamental function value. Fundamental means a long run relation to income, inflation and capital market interest rates. A market is conspicuous when its price level persistently deviates from the level explained by these variables.

The basic idea of the paper follows the literature on house price misalignments. Cuestas et al. examine European countries and analyse how far actual house prices deviate from prices that can be explained by fundamentals, as well as the real economic consequences of such deviations (Cuestas et al., 2023, p. 3215). The present approach transfers this logic to a separate country set and uses a Vector Error Correction Model (VECM) because the research question must explicitly separate the long run equilibrium relationship from short run adjustment. This separation is especially important for property markets because prices and income are typically trend driven.

1.2 Research Questions

The paper deliberately separates two methodologically different levels. First, it tests in a descriptive and econometric way whether a long run equilibrium relationship exists between property prices and macroeconomic fundamentals in 16 European countries. It also examines how the resulting misalignment series behave across countries and over time in terms of existence, shape and stability.

Building on this, the paper examines from a forecasting perspective whether the estimated VECM can generate usable pseudo out of sample forecasts of short run house price dynamics based on macroeconomic data. It also examines descriptively whether the Solid/Weak classification carries through to the path-error metrics, with the limitations of this examination spelled

out in §3 and §4.10. The methodological boundaries of this exercise are summarised in the limitations chapter.

A formal comparison of the VECM paths with simple time series benchmarks such as a first-order autoregressive process (AR(1)) or a random walk using the Diebold-Mariano test is not part of the scope of this paper and is proposed as a future research step. The transfer of misalignment signals into concrete investment strategies is also not considered.

1.3 Structure of the Report

After the introduction, the paper brings together relevant research strands on fundamentals of residential property prices, macroeconomic transmission mechanisms and the econometric logic of cointegration and error correction.

The hypotheses and the research design follow. This part defines the three testable hypotheses, explains the selection of countries, variables, data sources, transformations and diagnostic criteria, and specifies the construction of the misalignment signal as well as the design of the pseudo-out-of-sample backtest and the forward-looking projection.

The data analysis chapter moves from data harmonisation and unit-root tests to cointegration tests, the selection of preferred specifications and the country-by-country VECM diagnostics with crisis-dummy robustness.

The results chapter interprets model quality, long-run coefficients, adjustment parameters, the misalignment signal at the latest observation, the backtest performance and the forward-looking projection. The limitations section consolidates the methodological boundaries of the analysis in descending order of importance for the interpretation of the results.

The concluding chapter summarises the evidence on each hypothesis, places the empirical findings into the macroeconomic screening logic introduced in §2.4, and identifies the methodological extensions required to sharpen the forecast claims.

1.4 Relevance and Interest

The relevance of the topic follows from the dual role of home ownership. It is both a basic need and consumption good and a central component of wealth and collateral in the credit market. Case, Quigley and Shiller show that changes in housing wealth can have a relevant relationship

with consumption expenditure (Case et al., 2005, p. 1-3). Markets should therefore not only be assessed through asset quality and rent levels, but also through their macroeconomic embedding.

For the real estate industry, this analysis complements classical market observation. Transaction prices, rents, vacancy rates and yields remain decisive. However, they do not fully answer whether a market is cyclically overheated. A macroeconomic time series model can close this gap by showing whether price movements are supported by sustainable fundamentals. The European comparison also shows that similar price increases can have very different causes.

1.5 What to Expect

The expected outcome is not a ranking of buy or sell markets but a methodological grid. Three concrete expectations guide the analysis. First, a long run cointegration relationship between real house prices and the chosen fundamentals should be identifiable in most countries, because European housing markets share a common macroeconomic structure even where institutional details differ. Second, the diagnostic quality of these relationships should vary substantially across countries, so that a uniform Solid label across all 16 markets would itself be a warning sign of overfitting rather than a methodological success. Third, the short run forecast paths should remain economically close to the realised price path in the Solid markets, while the Weak markets should show systematically wider deviations or weaker direction sensitivity. The translation of misalignment signals into concrete investment decisions is not part of the scope.

2 Related literature

2.1 Fundamental values and misalignment

The research assumes that residential property prices are limited in the long run by economic variables such as income, financing costs, demographic demand, supply and institutional conditions. Girouard et al. place house price developments in Organisation for Economic Co-operation and Development (OECD) countries in relation to these fundamentals and stress that strong real price increases since the 1990s can only be sustainable when they are supported by demand, income and financing conditions. (Girouard et al., 2006, p. 9 ff.).

Geng extends this perspective by treating residential property prices in advanced economies as the result of long run structural and institutional drivers. This contribution is relevant for this

paper because it does not interpret valuation deviations as short term market mood, but as a potential signal of sustainability problems (Geng, 2018, p. 4 ff.).

The basic idea of Cuestas et al. is the closest point of reference. There, house price deviations are understood as the difference between actual prices and values that can be explained by fundamentals. The authors show that income is a central driver of fundamentally explainable house prices and that misvaluations can amplify business cycles in the short run, while overvaluations can dampen growth in the long run (Cuestas et al., 2023, p. 3215 f.). The present paper does not adopt the same model specification but follows the same basic logic. Price levels are interpreted economically only when they are placed in relation to a fundamental reference system.

2.2 Macroeconomic transmission channels

Residential property prices influence the real economy through several channels. Rising prices increase households' perceived wealth and thereby support consumption. This is especially important because home ownership is more widespread than equity ownership in many countries (Case et al., 2005, p. 4 ff.).

Through the credit channel, higher prices lead to higher collateral values and make debt financing easier, while falling prices tighten credit conditions. Higher prices also improve the expected margin of new construction projects and attract capital into project development. House price corrections can therefore weigh on consumption and investment (Girouard et al., 2006, p. 10. p. 28-32).

The variable selection follows this logic. Gross Domestic Product (GDP) per capita stands for purchasing power. Inflation represents the nominal price environment. The 10-year government bond yield represents financing costs and the return alternative. Property prices are the variable to be explained. Population was tested as a candidate variable, but the $I(1)$ filtering documented in §4.6 removes it from the system in all countries. The demographic demand component therefore remains outside the estimated model and is listed as a limitation in §6.8.

2.3 Econometric foundation

The econometric challenge is that property prices and many macroeconomic variables are not stationary. If trend driven series are regressed in levels, seemingly significant but economically unsound relationships can arise. Dickey and Fuller provide the classical test approach for unit

roots. Therefore, Augmented Dickey-Fuller (ADF) tests are first performed in levels and differences (Dickey & Fuller, 1979, p. 427 ff.).

Cointegration solves the problem of non-stationary variables when a linear combination is stationary. Engle and Granger show that cointegrated variables have an error correction representation, so that short run deviations react to long run disequilibria (Engle & Granger, 1987, p. 251 ff.). This is central for the research question because misvaluations are not only deviations from a model value. They are real disequilibria to which the system responds.

The Johansen approach extends this idea to multivariate systems and allows several cointegration vectors. Johansen develops likelihood-based tests for the cointegration rank in vector autoregressive (VAR) models (Johansen, 1991, p. 1551 ff). This paper therefore uses the Johansen trace test for each country and each price specification. If at least one cointegration rank is present, a VECM is estimated. This avoids interpreting long run coefficients from a system that statistically has no long run link.

2.4 Positioning of this paper

The paper positions itself between a valuation study and a macroeconomic forecast. It is not a pure replication because it uses its own quarterly panel of 16 countries up to 2025 and justifies the choice between real and nominal specification econometrically. At the same time, it is not an operative investment recommendation because it does not include asset level microdata, lease data or local supply restrictions. Its added value lies in a structured macroeconomic pre-selection that shows which markets are fundamentally conspicuous and where the statistical quality allows a stronger interpretation.

Unlike simple price income or price rent ratios, the VECM captures the joint dynamics of several variables. High prices can be fundamentally explainable when income, the inflation environment and the interest rate environment support them. Conversely, apparently cheap markets can appear risky when the fundamentals are also weak. This differentiation is more relevant for professional real estate decisions than one-dimensional valuation indicators.

3 Hypotheses

Table 1: Hypothesis overview

Hypothesis	Content	Empirical test
H1 Fundamental misvaluation	A long run cointegration relationship between residential property prices and the chosen macroeconomic fundamentals exists in the majority of the studied countries, and the estimated long run β coefficients are economically plausible in sign, namely positive on log GDP per capita and negative on the long-term yield. Deviations of observed prices from the equilibrium level implied by these fundamentals can therefore be operationalized as an economically interpretable misalignment series.	Cointegration test, sign and magnitude check of β coefficients in the preferred specification, deviation actual vs. fitted, contribution of long run adjustment.
H2 Model robustness	The model delivers consistent results and remains stable when specification, lags or crisis dummies are considered.	Stability checks, LM tests, Solid/Weak label, robustness check with GFC and COVID dummies.
H3 Out of sample consistency	The cointegration structures generate plausible medium term forecast paths. The primary criterion is the median MAE, which weighs all forecast quarters equally and is therefore robust to single quarter dislocations. The secondary criterion is the median RMSE, which is informative about path stability but is sensitive to individual large quarterly deviations such as the COVID shock 2020Q1 to 2020Q2 in the test window. Given the small group sizes ($n_{\text{Solid}} = 9$, $n_{\text{Weak}} = 6$) and the absence of a comparison against atheoretical benchmarks such as AR(1) or a random walk, H3 is not assessed as a hypothesis test in the inferential sense. It is treated as a descriptive consistency check between the in-sample diagnostic label and the out-of-sample path errors. A consistent ordering across both metrics would provide directional support; an inverted or noise-level ordering indicates that the diagnostic label does not separate path quality at this sample size and test design.	VECM by country until the end of 2017, 12 quarter forecasts, comparison of forecast vs. realized using RMSE and MAE, comparison of Solid vs. Weak.

H1 requires that actual prices deviate systematically from the prices expected from fundamentals and are not identified only through the significance of individual coefficients. H2 predicts that this deviation is not based on an unstable or diagnostically unsuitable specification. H3 reframes the in-sample diagnostic label as a question that can also be inspected out of sample. Given the constraints described in the H3 row above, the inspection is descriptive rather than inferential: it asks whether the Solid/Weak ordering carries through to the path-error metrics,

while remaining silent on whether any observed difference exceeds what sampling noise alone would produce.

The cautious wording reflects the heterogeneity of the European property market and the limited ability of macroeconomic models to directly capture local regulation, special tax regimes or institutional features. The expectation is that the country classification itself is a result of the analysis. Whether the Solid/Weak ordering also separates out-of-sample path quality is treated as an open empirical question rather than a directional prediction.

4 Research Design

4.1 Research Strategy

Instead of pooling all countries in one panel model, a separate system is estimated for each country. European countries differ too strongly in ownership rates, credit structures, rent regulation, construction elasticity, demography and inflation history for a pooled model to represent this heterogeneity meaningfully. A country specific VECM allows both the long run relation and the speed of adjustment to vary by market.

The study period generally runs from 1990Q1 to the latest available observations in 2025. This length is necessary because the model is intended to identify long run equilibrium relationships and therefore needs several interest rate, inflation and business cycles as well as crisis episodes. For individual countries, however, the effective sample begins later because BIS coverage does not reach back to 1990 everywhere.

The country selection includes Austria, Belgium, Switzerland, Germany, Denmark, Spain, Finland, France, the United Kingdom, Greece, Hungary, Italy, the Netherlands, Norway, Portugal and Sweden.

This selection is economically meaningful because it includes large core markets, smaller open economies, Nordic markets, Southern European markets and transformation examples. It therefore does not create a homogeneous fair-weather dataset, but a comparison with deliberately heterogeneous market structures.

4.2 Data Collection

Data collection follows the principle of using data sources that are as standardised, public and replicable as possible. Residential property prices come from the Residential Property Price Index (RPPI) of the Bank for International Settlements (BIS) and are obtained through the Federal Reserve Economic Data (FRED) database. The index is available for nominal and real prices. GDP per capita is taken from FRED as a real per capita measure. Inflation is derived from FRED consumer price index (CPI) data. Population comes from the World Bank series Population, total. The 10-year government bond yield is obtained from FRED as the long-term benchmark yield.

The choice of these sources is part of the methodology. For a European country comparison, consistency is more important than maximum national detail. National databases could sometimes provide longer or finer series, but they would not necessarily be harmonised. FRED and the BIS series allow a uniform data logic, and World Bank population provides an internationally comparable demographic basis. This makes the analysis replicable and less vulnerable to selective data choice. Two specific issues concern the GDP series. For 14 countries, the Eurostat consistent series CLVMNACSCAB1GQ{XX} (Chain Linked Volumes, National Accounts) is used. For Greece, the Eurostat country code 'EL' is used instead of the International Organization for Standardization two-letter (ISO 2) code 'GR'. Switzerland uses the almost identical variant CLVMNACSAB1GQCH, which is seasonally adjusted but not calendar adjusted. The missing calendar adjustment is numerically irrelevant for log level and log difference relationships. For the United Kingdom, the FRED series NGDPRSAXDCGBQ (Real GDP, seasonally adjusted, in national currency in chained prices) is the closest equivalent. All three sources measure real GDP in a national chained price logic. The methodological details differ, but for the log differences and log level relationships used here, this heterogeneity is numerically small. It remains a minor limitation of the data harmonisation. For the 10-year yield, the monthly FRED series IRLTLT01{XX}M156N is used primarily for each country.

All series are aligned to quarterly frequency with period start. Monthly yields are aggregated to quarters by taking simple averages. This decision is deliberately conservative. An average of the monthly values describes the interest rate environment of a quarter better than a single month end value. Population is available annually and is the only series interpolated log linearly, that is geometrically, to quarters.

Table 2: Overview of macro factors

Variable	Source	Processing	Economic function
Property prices	BIS Residential Property Price Index via FRED	Nominal and real, index 2010 = 100, quarterly data	Central valuation variable and target variable
GDP per capita	FRED national accounts	Real per capita measure, logged	Purchasing power and income as the demand base
Inflation	FRED consumer prices	Percentage value in levels	Nominal environment and price stability
Ten-year yield	FRED long term government bond yield	Monthly to quarterly average, level	Financing costs, discounting and return alternative

4.3 Data Cleaning

Data cleaning begins with a uniform country identifier. All countries are managed using ISO 2 codes so that series from different sources can be linked unambiguously. Variable names follow a fixed scheme: Property Real, Property Nominal, GDP per capita, Inflation, Pop and Yield 10Y. This standardisation may appear technical, but it is methodologically important. It reduces the probability that country series are linked incorrectly or that specifications are processed differently.

Property prices, GDP per capita and population are logged. This makes growth rates easier to interpret and reduces extreme level differences between countries. Inflation and the 10-year yield remain in levels because they are already percentage variables. This distinction prevents percentage points from being artificially logged. As the house price specification, we use the real BIS index deflated by the national consumer price index. The reason for this choice is explained in §4.5. The nominal variant is estimated in parallel and reported in Appendix G as a robustness comparison, but it is not used for the main results.

Missing observations are not hidden through broad interpolation. If property prices, GDP per capita or yields are missing, the corresponding observation is not artificially created. This rule is decisive because interpolation would smooth precisely the dynamics that a VECM is intended to explain. Only population is interpolated. Specifically, annual values are logged, linearly filled to quarters and then transformed back. This log linear approach treats population as a geometrically growing variable and assumes a constant growth rate between two annual observations. The series is used only for ADF diagnostics. It does not meet the $I(1)$ filter condition in any country and is consistently removed before estimation (see §4.6).

4.4 Data Presentation

The exploratory figures in Appendix B serve two purposes. First, they show whether the datasets develop in an economically plausible way. Second, they prepare the later model interpretation. GDP per capita shows long run upward trends in most countries, but with visible breaks during crisis periods. Inflation is clearly more volatile and shows country specific shocks. The property price series show especially strong persistence and different cycle patterns.

Nominal prices rise strongly in many countries, but real prices show whether this increase goes beyond general price developments. In an inflationary environment, a nominal price increase can be much less important in real terms. For this reason, and because of the econometric considerations on the integration structure of the system discussed in §4.5, we use the real house price series consistently as the dependent variable.

The yield figures show a long run decline in capital market interest rates, which is especially relevant for property markets. Falling yields mechanically increase the present value of future cash flows and make property more attractive relative to bonds. At the same time, very low interest rates can favour price exaggerations. The 10-year government bond yield is therefore included as the long-term capital market reference rate and is not replaced by short term policy rates.

4.5 House price specification: real, not nominal

As the dependent variable, we use the real BIS house price index deflated by the national consumer price index. This choice follows an econometric, an economic and a literature-based consideration. It is not the result of choosing between the real and nominal specification based on Solid Weak diagnostics. It is part of the model justification. The only exception is the status Not Suitable, defined as $\max. \text{AR root} > 1.01$. Such estimates are generally rejected.

Econometrically, this choice is necessary. By construction, the nominal house price contains the accumulated price level. The log nominal house price equals the log real house price plus the log consumer price index.

A cointegration system with the nominal house price therefore implicitly combines a real house price component, typically $I(1)$, with an accumulated price level component, also $I(1)$. The inflation rate used in the model measures the change. As a result, the accumulated inflation level component of the nominal house price must be absorbed by the estimated β coefficients of the

other $I(1)$ variables. The coefficient of log GDP per capita performs two tasks at the same time, namely the economic fundamental effect and the hidden absorption of inflation drift.

Economically, the research question focuses on the deviation of the real market valuation from the real fundamental value. A formal translation of the misalignments into return expectations is outside the scope of this paper. Forecast performance and conditional paths are documented in §6.4 and §6.6 without a return calculation.

From a literature perspective, the choice follows common practice. Cuestas et al. (2023), Geng (2018) and Girouard et al. (2006) consistently use real prices in their analyses of European and OECD house prices. A deviation from this convention would require justification, not its adoption.

For countries where the Johansen test in the real specification does not deliver a cointegration rank, we report the nominal estimate as a fallback and mark these countries accordingly in the results tables. Specifically, this concerns Hungary and Sweden. In both cases, the real variant does not provide a VECM eligible system, which is why the nominal estimate is used for these two countries. For the other 13 real specified countries in the dataset, the real specification is used.

Before the specification decision, an $I(1)$ filter rule is applied. It tests the macroeconomic candidate variables GDP, inflation, population and yield for the $I(1)$ property and removes $I(0)$ variables from the system. The house price variable is always retained as the target variable. A country is de facto excluded from the VECM sample if, after this filtering, fewer than two variables remain or if the subsequent Johansen trace test in both specifications, real and nominal, delivers no cointegration rank. This specifically concerns Denmark, which is therefore removed from the evaluable sample.

This asymmetry is made transparent in the data analysis and results chapters and is considered when the misalignment series are interpreted.

4.6 Data Analysis Method

The first analytical step is the ADF tests. They examine whether the respective series are stationary in levels or whether only the first difference becomes stationary.

The second step is the Johansen cointegration test. For each country and each price specification, real and nominal, a system with up to five variables is tested. Its actual dimension by country follows from the ADF based $I(1)$ filtering (see the following paragraph and §4.5). The Johansen trace test is used. The cointegration rank is determined based on the trace statistic at a significance level of five percent. A VECM is estimated only if at least one rank is present. Models enter the cointegration test only if the jointly available sample covers at least 40 quarters, which corresponds to ten years. This lower bound secures reliable lag selection and avoids relying on the asymptotic properties of the tests in series that are too short. For the subsequent VECM estimation and the robustness check with crisis dummies, a slightly lower effective threshold of 35 quarters is used because lag differencing in the VECM consumes observations. In practice, no country moves from the cointegration test into the VECM estimation with fewer than 36 effective quarters.

The number of endogenous variables per country is not fixed. Before estimation, the ADF test is applied to every candidate variable in each country. Only variables whose level series is not stationary and whose first difference becomes stationary, meaning $I(1)$ candidates, are included in the system. The house price series is always included as the target variable. This strict $I(1)$ filtering prevents $I(0)$ variables from artificially distorting the estimated cointegration rank. The price of this rule is that the effective system dimension varies by country between two variables, for example Finland with HP and GDP, and four variables in the full system. In practice, the filtering affects the logged population series in every one of the 16 countries. $\log \text{Pop}$ fails both level and first difference stationarity at the 5 percent level in all countries ($\text{lev_p} \in [0.145; 0.986]$; $\text{diff_p} \in [0.074; 0.889]$) and therefore meets the $I(1)$ filter condition nowhere. The consistent lack of stationarity in the first difference is compatible with an integration order of $I(2)$, but in the present setup we cannot distinguish $I(2)$ from a deterministic trend stationary process with strong persistence at the chosen sample length. Either reading is sufficient to remove $\log \text{Pop}$ from the $I(1)$ system used here. For this reason, $\log \text{Pop}$ is removed from the system in all countries. The specific variable selection by country is shown in Table 3 (§5.3).

The $I(1)$ filter rule defined above is applied only to the macroeconomic candidate variables. The house price series is always included in the system as the target variable by construction, regardless of its ADF finding. This special treatment must be documented. For seven countries, the ADF tests do not classify the real house price series as an $I(1)$ candidate. ES Real is rejected

at the level as trend stationary ($\text{lev}_p = 0.043$). FR Real, GB Real, GR Real, HU Real and NL Real fail stationarity in the first difference. DK Real fails both tests and is in any case removed from the sample because of the filter logic for the remaining variables. The econometric consequences are taken up in the limitations chapter.

The third step is the VECM estimation. The VAR lag length is selected by the `statsmodels select_order` function using the Akaike Information Criterion (AIC) over a fixed search range from 1 to 6 lags. The function returns the lag index corresponding to the minimum AIC value. A technical minimum of $k_{ar_diff} = 1$ (corresponding to a VAR lag length of 2) is enforced. If the lag selection encounters a numerical exception, a default of $k_{ar_diff} = 2$ is applied.

For Germany and France, both in the real specification, the upper search bound is binding. The AIC selection delivers $k_{ar_diff} = 5$, corresponding to a VAR lag length of 6. Extending the search range could produce a higher number of lags for these two countries. A sensitivity check with an extended search grid is listed as a follow up step in §6.8.

4.7 Economic Workflow

The estimation and diagnostic workflow follow a fixed pipeline. It consists of data collection from BIS, FRED and the World Bank, cleaning and logging of the level series, ADF based $I(1)$ filtering, Johansen trace tests for each country specification, VECM estimation with AIC lag selection, LM and eigenvalue diagnostics, crisis dummy robustness for the models classified as Solid, and backtesting plus forward projection. Each step is automated in Python and persisted in its own CSV table so that the pipeline remains traceable step by step.

4.8 Model diagnostics and quality labels

Residual autocorrelation is tested with Lagrange Multiplier (LM) tests at equation level. Stability is assessed through the roots of the VAR representation. Because a VECM contains integrated variables, a root close to one is expected and is not automatically an error. This distinction prevents a correct cointegration model from being falsely rejected as unstable. The LM test is carried out equation by equation. For each model, the minimum p value across all k equations is evaluated. To keep the family wise error rate at 5 percent for k equations, the Solid threshold is Bonferroni corrected to $0.05 / k$. This correction is especially relevant for the reduced bivariate and trivariate specifications, where the nominal error rate would otherwise be inflated.

A model receives the label Solid if the minimum equation level LM p value meets the family wise Bonferroni threshold of $0.05/k$ and the maximum absolute eigenvalue of the VAR companion matrix does not exceed $1 + 10^{-6}$, which is the numerical tolerance for the VECM unit root caused by construction. Models that fail at least one of these two criteria, but whose maximum eigenvalue modulus remains below 1.01, receive the label Weak. Models with a maximum eigenvalue modulus above 1.01 are rejected as Not Suitable. The specification choice is justified in §4.5 and is set uniformly to real for all countries. The labels therefore describe only the diagnostic quality of the selected real estimate and determine the confidence of interpretation. The threshold of 1.01 is deliberately narrow. An eigenvalue modulus of 1.04 would already generate a five-fold shock amplification over a 10-year horizon and would therefore no longer be numerical noise.

A Weak model can be economically plausible, but its short run dynamics are statistically less clean. For the research question, this means that H1 can also receive evidence in Weak countries, but H2 is only fulfilled to a limited extent there, and H3 must be assessed more cautiously in Weak countries.

4.9 Construction of the misalignment signal

The central evaluation object derived from the country specific VECM is the misalignment signal. It is the deviation of the observed log house price from the level implied by the estimated long run cointegration relationship and is therefore based on the β logic of the equilibrium structure, not on the α lag dynamics of the full VECM.

Let Y_t denote the country specific vector of endogenous $I(1)$ variables, with the log real house price as its first element. Let β denote the matrix of estimated cointegration vectors with rank $r \geq 1$. The cointegration deviation is defined as

$$u_t = \beta' Y_t - \mu,$$

where μ is the mean of $\beta' Y_t$ over the estimation sample, so that u_t is centred at zero by construction. Each column of u_t represents one identified long run relationship.

The misalignment in the house price dimension is the projection of u_t onto the house price equation,

$$m_t^{\text{HP}} = \log \text{HP}_t - \log \text{HP}_t^{\text{eq}},$$

where $\log HP_t^{eq}$ is the equilibrium log house price implied by β when the remaining $I(1)$ variables are held at their observed contemporaneous values. A positive m_t^{HP} indicates that observed prices lie above the level explained by income, inflation and yields. A negative m_t^{HP} indicates the opposite. By construction, the sample mean of m_t^{HP} over the estimation window is zero, so the signal measures relative over and undervaluation against the country specific long run reference, not an absolute valuation level. If a country has been persistently above or below its true unobserved equilibrium throughout the estimation window, for example because of a long-run decline in equilibrium real interest rates or a structural credit-supply expansion not captured by the modelled yield series, the misalignment series cannot detect this offset; it is absorbed into the centred reference itself and reported as an average of zero. Cross-country comparisons of the magnitude of m_t^{HP} must be read with this asymmetry in mind.

The economic identification of m_t^{HP} is examined in §6.2 through the pure beta fit in Figure H3, which removes the α lag dynamics and shows whether the estimated long run relationship alone carries the level path. Two complementary adjustment visualisations support this reading. Figure H1 shows the 1 step ahead fit of log house prices, including the α correction term and Γ lag dynamics. Figure H2 shows the same fit in first differences. If the pure beta path in Figure H3 systematically matches the realised levels in direction and magnitude, the requirement for the economic interpretability of the misalignment series, and therefore of the out of sample paths in §6.4 and the forward projection in §6.6, is met

For the six countries with full cointegration rank (ES, FI, GB, NL, NO, SE), the individual β coefficients depend on the chosen normalisation and identification scheme. In these cases, m_t^{HP} is still well defined as the deviation of the observed price from the equilibrium implied level, but the decomposition of this deviation into individual fundamental contributions must be read with caution. This restriction is taken up as a limitation in §6.8.

4.10 Pseudo-Out-of-Sample-Validation

The estimation and diagnostic chain described in §4.6 to §4.8 evaluates the VECM purely in sample. The test follows a fixed training and test window design. The training cutoff is set to the fourth quarter of 2017. For each country, all observations up to and including this date are assigned to the training sample, while all later observations are assigned to the test sample. The VECM is estimated on the training sample with the same specification as the main model from §4.6.

From the system estimated in this way, a closed multi step forecast is generated for the following twelve quarters. Closed means that no realised test sample values enter the calculation during this forecast horizon. The model continues its own forecast quarter by quarter and relies exclusively on its own estimated parameters and its own previously generated forecast values.

Four methodological restrictions of the test must be made transparent. First, the $I(1)$ filtering of the endogenous variables is calculated once on the full available sample and then applied to both the main estimation and the backtest. A refiltering restricted to the training sample could yield slightly different variable sets in individual countries. Second, the VAR lag length selected by AIC is taken from the full sample and is not re-selected on the training sample alone. Third, the cointegration rank from the Johansen trace test is fixed from the full sample. Fourth, the ADF-based $I(1)$ classification and the Johansen rank determination are computed from a single pipeline run on the complete dataset; they are not recalibrated for each possible training window. These four transfers mean that the backtest does not constitute a clean out-of-sample test in the strict sense. Hyperparameters that should ideally be reselected on the training window alone are taken from the full sample, which mechanically biases the path errors downward relative to a fully recalibrated design. The reported RMSE and MAE values should therefore be read as a lower bound on the true out-of-sample path error, not as an unbiased estimate of it. We acknowledge this as a methodological limitation and discuss it jointly with the other backtest restrictions in the limitations chapter.

Forecast quality is measured for each country through two path error metrics. The first is the root mean squared error, RMSE. The second is the mean absolute error, MAE. Both are based on the difference between predicted and realised log house prices. Both metrics are reported in units of the log level. A value of about 0.10 therefore corresponds approximately to a cumulative path deviation of ten percent relative to the realised house price path. RMSE penalises large individual deviations disproportionately and is therefore sensitive to quarters with especially high forecast errors. MAE weights all quarters equally and is therefore more robust to individual outliers. Reporting both measures in parallel makes it possible to see whether a path error is distributed evenly across the horizon or dominated by individual quarters.

The effective training length varies by country with the beginning of the available BIS series. It ranges from 19 years for Hungary, with BIS coverage starting in 1999Q1, to 28 years for Finland, with BIS coverage starting in 1990Q1. The shorter training samples in Hungary, 19 years, Greece,

about 21 years, and Italy, 22 years, should be read as an additional source of estimation uncertainty for the path errors of these countries and are contextualised accordingly in §6.4. The estimated cointegration relationship is therefore not identified only by a later crisis phase, but by the full pre crisis and crisis history. The test sample covers the quarters 2018Q1 to 2020Q4 and therefore includes the COVID shock in 2020Q1 to 2020Q2. This asymmetry follows from the chosen fixed 12 quarter length and is not the result of a differentiated shock selection.

The choice of a forecast horizon of twelve quarters follows two considerations. First, a 3-year horizon corresponds to the time scale over which house price adjustments typically take place and is therefore the economically relevant observation window for early warning indications (Girouard et al., 2006, p. 28 ff.). Second, at very long horizons, a VECM tends toward mean reversion. The dynamics are then dominated by the estimated cointegration constant, and the paths become flat. From about 16 quarters onward, this mechanical property becomes so strong that path errors essentially measure only the difference between the realised trajectory and the estimated unconditional mean line. The horizon chosen here is deliberately below this threshold.

4.11 Forward-looking Projection

The out of sample test in §4.10 serves model validation. A second application of the estimated VECM, clearly separated from it, is the forward-looking projection. It is a conditional path of log house prices into the future, given the identified equilibrium structure. Unlike in §4.10, no training holdout is used here. Instead, the VECM is estimated on the full available dataset for each country, and the forecast starts immediately after the last observed quarter and extends to the fourth quarter of 2026. The forecast horizon is therefore country specific. Because BIS coverage and the FRED series end at different points in time for the individual countries, each country has a different number of forecast quarters. This number is exactly the difference between the last available data point and the target quarter at the end of 2026.

In addition to the point forecast, pointwise confidence bands at a level of 90 percent are reported. These bands are derived from the estimated forecast covariance of the model and are placed around the point forecast quarter by quarter. They represent the expected dispersion of future log house prices under the estimated model parameters. The band widens as the horizon increases because forecast uncertainty in a dynamic system builds up over time. A narrow band at the endpoint indicates a country with a relatively stable estimated adjustment and shock

structure. A wide band indicates large model side uncertainty. The pointwise 90 percent bands are derived from the asymptotic forecast covariance matrix of the VECM. The underlying assumption is that the estimated cointegration coefficients and the estimated adjustment and short run parameters remain valid during the forecast period. They are not risk bands against structural breaks.

The forward-looking projection is explicitly conditional. It projects the path that would result if the cointegration relationship estimated in the full sample and the estimated adjustment and short run parameters continued to hold. It is not a stress test and not a structural break model. No explicit inflation or interest rate shock is modelled. Possible shocks operate only to the extent that they arise mechanically from the estimated reaction functions of the system. This restriction is part of the conditional reading of the projection.

5 Data Analysis

5.1 Exploratory patterns

The exploratory analysis first shows that the countries examined do not share one uniform European property price development. Some markets, such as Germany, Switzerland, Sweden or Norway, show strong nominal upward trends over long periods. Other markets, such as Greece, Spain or Italy, contain more distinct crisis and recovery phases. This heterogeneity confirms the decision to estimate country specific systems.

GDP per capita shows a long run trend that is interrupted in crisis years. The financial crisis from 2008, the European sovereign debt phase and the COVID period leave visible traces in several countries. Inflation, by contrast, shows shorter and stronger movements. The period after 2021 is especially conspicuous in many countries. Population develops more smoothly, but it can provide important demand information in countries with migration, ageing or population decline.

The yields on 10-year government bonds fall over the long run in almost all countries. From a property perspective, this decline is a central valuation impulse because lower discount rates can justify higher prices. At the same time, the decline in interest rates can hide overvaluations if investors extrapolate low return requirements permanently. Even before estimation, the

exploratory analysis therefore shows that price development would be interpreted incompletely without interest rates and demography.

5.2 Stationarity and cointegration

The ADF tests confirm the expectation that many series are not stationary in levels and become more stable only in first differences. This is exactly why a simple Ordinary Least Squares (OLS) regression of price levels on macroeconomic levels would be problematic. It could show high explanatory power because all series are trend driven, without a reliable equilibrium relationship. The unit root test is therefore not only a formal statistical step, but also protection against spurious correlation.

Of the 15 VECM eligible countries, 13 show cointegration support already in the real specification. Hungary and Sweden fall back to the nominal variant; the rationale for this fallback rule is given in §4.5. Across all 15 countries, the specification reported for each country therefore contains a cointegration relationship. This overall result is strong evidence for H1 because it means that property prices and the selected fundamentals do not move independently from each other in the long run.

Table 3: Lag Length Selection (AIC, Max 6 Quarters)

Country	Spec	Endogenous variables	Sample	N	k _Δ	Rank	VECM
AT	Nominal	HP_N;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	1	2	Yes
AT	Real	HP_R;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	1	2	Yes
BE	Nominal	HP_N;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	3	3	Yes
BE	Real	HP_R;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	3	1	Yes
CH	Nominal	HP_N;GDP_pc;Yield	1995Q1–2025Q1	121	1	1	Yes
CH	Real	HP_R;GDP_pc;Yield	1995Q1–2025Q1	121	1	1	Yes
DE	Nominal	HP_N;GDP_pc;Inflation;Yield	1991Q1–2025Q1	137	5	4	Yes
DE	Real	HP_R;GDP_pc;Inflation;Yield	1991Q1–2025Q1	137	5	3	Yes

DK	Nomi- nal	HP_N;GDP_pc;Yield	1995Q1– 2025Q1	121	3	0	No
DK	Real	HP_R;GDP_pc;Yield	1995Q1– 2025Q1	121	3	0	No
ES	Nomi- nal	HP_N;GDP_pc;Inflation	1995Q1– 2025Q1	121	4	3	Yes
ES	Real	HP_R;GDP_pc;Inflation	1995Q1– 2025Q1	121	4	3	Yes
FI	Nomi- nal	HP_N;GDP_pc	1990Q1– 2025Q1	141	2	2	Yes
FI	Real	HP_R;GDP_pc	1990Q1– 2025Q1	141	2	2	Yes
FR	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	5	1	Yes
FR	Real	HP_R;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	5	2	Yes
GB	Nomi- nal	HP_N;GDP_pc;Inflation	1991Q1– 2025Q1	137	4	3	Yes
GB	Real	HP_R;GDP_pc;Inflation	1991Q1– 2025Q1	137	4	3	Yes
GR	Nomi- nal	HP_N;GDP_pc;Yield	1997Q2– 2025Q1	112	1	1	Yes
GR	Real	HP_R;GDP_pc;Yield	1997Q2– 2025Q1	112	1	1	Yes
HU	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1999Q1– 2025Q1	105	1	1	Yes
HU	Real	HP_R;GDP_pc;Inflation;Yield	1999Q1– 2025Q1	105	4	0	No
IT	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1996Q1– 2025Q1	117	3	4	Yes
IT	Real	HP_R;GDP_pc;Inflation;Yield	1996Q1– 2025Q1	117	3	2	Yes
NL	Nomi- nal	HP_N;Inflation;Yield	1991Q1– 2025Q2	138	3	3	Yes
NL	Real	HP_R;Inflation;Yield	1991Q1– 2025Q2	138	3	3	Yes
NO	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	4	4	Yes
NO	Real	HP_R;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	4	4	Yes
PT	Nomi- nal	HP_N;GDP_pc;Yield	1995Q1– 2025Q1	121	1	2	Yes

PT	Real	HP_R;GDP_pc;Yield	1995Q1–2025Q1	121	1	1	Yes
SE	Nominal	HP_N;GDP_pc;Yield	1993Q1–2025Q1	129	1	3	Yes
SE	Real	HP_R;GDP_pc;Yield	1993Q1–2025Q1	129	4	0	No

Note. Spec = real or nominal house-price specification. Endogenous variables: HP_R = log real house price, HP_N = log nominal house price, GDP_pc = log GDP per capita, Inflation = CPI YoY %, Yield = 10Y bond yield. k_{Δ} is the lag order in differences chosen by AIC (max 6). Rank is the cointegrating rank from the Johansen trace test at the 5% level. "VECM = Yes" requires Rank ≥ 1 and $N \geq 40$.

Denmark is the systematic exception and is removed from the sample before the VECM estimation. In the Danish dataset, the ADF test with variable specific trend and constant choice identifies several candidate variables, inflation and the 10-year yield, as already stationary. In the real specification, after $I(1)$ filtering, Property_Real, GDP per capita and Yield remain as three non-stationary candidate variables. One further result from the $I(1)$ filtering warrants mention: for the Netherlands, the ADF test classifies log GDP per capita as an $I(0)$ variable in both specifications, so it is removed from the Dutch system before the Johansen test. The subsequent Johansen trace test on these 3 variable systems delivers a cointegration rank of 0 at the 5 percent significance level. No rank is identified in the nominal specification either. For Denmark, the empirical prerequisite for an identifiable cointegration relationship for VECM estimation is therefore missing in both variants. We therefore consistently refrain from specifying a VECM for Denmark and reduce the evaluable country sample from the original 16 to 15. The economic interpretation of this omission is that the fundamental drivers in the Danish sample may not form an identifiable common equilibrium relationship with real house prices, possibly because of the stabilisation policy of the Danish krone against the euro and the longer disinflation period. This makes a misalignment analysis within the VECM framework materially irrelevant for Denmark.

5.3 Preferred specifications

The choice of the real house price specification justified in §4.5 applies consistently to all countries. There is no longer a selection between the real and nominal variant based on diagnostics. For Hungary and Sweden, where the Johansen test in the real specification does not deliver a cointegration rank, the estimate falls back to the nominal variant and is marked accordingly in the results tables.

For Hungary, it must additionally be noted that the nominal β coefficients of the fallback estimate ($\beta_{\text{GDP}} = -22.205$, $\beta_{\text{Infl}} = 1.568$, $\beta_{\text{Yield}} = -2.775$) are of a magnitude that points to a cointegration vector close to the numerical identification boundary. We do not read these values as structural elasticities, but as an estimated linear combination that just identifies the level relationship in the 26-year sample (1999Q1 to 2025Q1). This sample starts after the beginning of Hungary's disinflation and convergence phase and excludes the hyperinflation episode of the early 1990s. Its explanatory power is therefore reduced. The Hungarian misalignment series is read with increased caution in §6.6 and §6.8.

A separate identification caveat applies to the six countries for which the Johansen test selects a cointegration rank equal to the system dimension (ES, FI, GB, NL, NO, SE). When the rank equals the dimension, no linear combination of the system variables remains non-stationary, and the individual β coefficients of the first cointegration vector primarily reflect the identification convention of the estimator rather than an identified structural relationship. The misalignment series for these countries remains technically defined as the projection of the deviation onto the house-price equation, but its economic content rests primarily on the adjustment dynamics α and on the joint movement of the system, not on the individual β -vector. This restriction also applies to the cross-country β -comparison in Figure H10 and is taken up again in §6.8.

According to the diagnostic results, 9 of 15 countries are classified as Solid: Austria, Switzerland, Germany, Spain, Finland, the United Kingdom, Hungary, Norway and Sweden, with Hungary and Sweden in the nominal variant. 6 countries are classified as Weak: Belgium, France, Greece, Italy, the Netherlands and Portugal. Compared with the earlier nominal variant, the distribution shifts toward more Weak classifications among the real specified countries. All preferred models are stable in terms of the eigenvalue diagnostics. The Weak classification does not arise from instability, but from residual autocorrelation. This distinction is methodologically important because Weak does not mean unusable. It signals that the short run dynamics are captured incompletely and that the results must be interpreted more cautiously. Three Solid countries, the United Kingdom, Norway and Sweden, pass the Bonferroni corrected LM test with LM p values only slightly above the country specific threshold $0.05/k$ (GB: 0.0240 vs. 0.0167 at $k=3$; NO: 0.0156 vs. 0.0125 at $k=4$; SE Nominal: 0.0362 vs. 0.0167 at $k=3$). They are therefore Solid on the same diagnostic path as the other six countries in this class, although the margins are smaller.

5.4 VECM interpretation

The VECM results are read through beta and alpha. Beta describes the long run equilibrium relation. Because the house price coefficient in the first vector is normalised to one, the other coefficients are not simple elasticities in the classical regression sense. Their sign and magnitude show which variables dominate the long run relation in the respective country. When there are several cointegration vectors, it is especially important not to overinterpret every individual zero value in the first vector.

Table 4: VECM Diagnostic Results – LM Autocorrelation Test and Eigenvalue Stability

Country	Spec	k_eq	Bonf. thr.	LM p (min)	No auto-corr.	Max root	Stable	Evaluation
AT	Nominal	4	0.0125	0.1867	Yes	1.0000	Yes	Solid
AT	Real	4	0.0125	0.2253	Yes	1.0000	Yes	Solid
BE	Nominal	4	0.0125	1.153e-06	No	1.0000	Yes	Weak
BE	Real	4	0.0125	1.030e-06	No	1.0000	Yes	Weak
CH	Nominal	3	0.0167	0.1382	Yes	1.0000	Yes	Solid
CH	Real	3	0.0167	0.2137	Yes	1.0000	Yes	Solid
DE	Nominal	4	0.0125	0.2567	Yes	0.9909	Yes	Solid
DE	Real	4	0.0125	0.2791	Yes	1.0000	Yes	Solid
ES	Nominal	3	0.0167	0.0486	Yes	0.9769	Yes	Solid
ES	Real	3	0.0167	0.2835	Yes	0.9829	Yes	Solid
FI	Nominal	2	0.0250	0.0570	Yes	0.9777	Yes	Solid
FI	Real	2	0.0250	0.1481	Yes	0.9525	Yes	Solid
FR	Nominal	4	0.0125	0.0016	No	1.0000	Yes	Weak
FR	Real	4	0.0125	0.0113	No	1.0000	Yes	Weak
GB	Nominal	3	0.0167	0.0965	Yes	0.9607	Yes	Solid
GB	Real	3	0.0167	0.0241	Yes	0.9580	Yes	Solid
GR	Nominal	3	0.0167	0.0227	Yes	1.0000	Yes	Solid
GR	Real	3	0.0167	9.717e-08	No	1.0000	Yes	Weak
HU	Nominal	4	0.0125	0.1953	Yes	1.0000	Yes	Solid
IT	Nominal	4	0.0125	0.0025	No	0.9491	Yes	Weak
IT	Real	4	0.0125	0.0073	No	1.0000	Yes	Weak
NL	Nominal	3	0.0167	4.358e-05	No	0.9813	Yes	Weak
NL	Real	3	0.0167	2.258e-05	No	0.9745	Yes	Weak

NO	Nominal	4	0.0125	0.0119	No	0.9830	Yes	Weak
NO	Real	4	0.0125	0.0156	Yes	0.9760	Yes	Solid
PT	Nominal	3	0.0167	0.4066	Yes	1.0000	Yes	Solid
PT	Real	3	0.0167	1.760e-05	No	1.0000	Yes	Weak
SE	Nominal	3	0.0167	0.0362	Yes	0.9851	Yes	Solid

Note. α_{HP} is the speed-of-adjustment loading on the house-price equation. Negative values indicate that house prices themselves move toward the long-run equilibrium following a deviation; positive or near-zero values indicate that adjustment occurs primarily through other system variables. Blank cells indicate variables removed by the $I(1)$ filter.

Alpha describes adjustment to the long run disequilibrium. A negative α value in the house price equation points to direct equilibrium correction through prices. In the preferred specification for each country, the following picture emerges, sorted by the algebraic α value from strongly negative to strongly positive. Norway shows the clearest negative house price loading with $\alpha = -0.0886$, followed by Finland (-0.0585), Switzerland (-0.0534) and the United Kingdom (-0.0445). France (-0.0369), Spain (-0.0348), Sweden (-0.0300 in the nominal variant) and Austria (-0.0292) show a medium adjustment speed. The house price loadings in the Netherlands (-0.0135), Belgium (-0.0052) and Italy (-0.0013) are small to negligible. Three countries show an economically interpretable positive α in the house price equation: Greece ($+0.0388$), Portugal ($+0.0050$) and Germany ($+0.0028$). The Hungarian fallback estimate delivers a nominal α of $+0.0012$, which is close to the numerical identification threshold and is read economically as practically zero. The following discussion therefore focuses primarily on the three real specified countries with positive α . In economic terms, this means that direct price adjustment in these three countries does not lead back toward equilibrium. The mean reversion argument runs there through other variables of the system. In Germany and Italy, the direct price channel is numerically almost switched off. Adjustment must occur through the GDP, inflation or yield equations. In Greece and Portugal, the positive sign is an indicator that the estimated long run relation holds without supporting adjustment dynamics in the house price equation. This is a feature that must be read cautiously in the forecast evaluation (§6.4) and methodologically adds to the Weak classification. A second reading of these positive α_{HP} estimates should be acknowledged. For DE ($+0.0028$) and PT ($+0.0050$) the magnitudes are close enough to zero that the sign is not robustly identified given the present sample lengths and Bonferroni-corrected thresholds. The economic interpretation used here, namely that the price-side channel is inactive and the system equilibrates through the other equations, therefore cannot be cleanly separated from a mild misspecification of the cointegration vector that flips the sign of α_{HP} at near-zero

magnitudes. We retain the active interpretation but flag this ambiguity for DE, PT and the Hungarian fallback case.

5.5 Robustness checks

The robustness check re estimates those models that were classified as Solid according to the diagnostic criteria in §4.8 with exogenous crisis dummies for the Global Financial Crisis (GFC; 2008Q1 to 2009Q4) and COVID-19 (2020Q1 to 2021Q4). Rank and lag structure are taken from the dummy free baseline test and are not identified again. The test therefore examines the stability of the α estimate in the house price equation when controlling for exogenous crisis periods, not the sensitivity of the rank determination itself.

The scope of the robustness check does not fully match the list of preferred specifications from §5.3. In AT, CH, DE, ES, FI and GB, both the real and the nominal specification are classified as Solid. For these countries the robustness check is run on both Solid variants and reported in Table F1, with the real variant as the preferred reference and the nominal variant as the parallel comparison. Table F1 lists both specifications for AT, CH, DE, ES, FI and GB; for GB the entry follows the same logic, with the real variant as the preferred and tested reference. Norway is Solid in the real specification, while the nominal variant does not meet the Solid criterion, and is therefore tested on the real variant, which is both preferred and the only Solid variant.

In two countries, GR and PT, only the nominal specification is Solid, while the real variant reported as preferred in §5.3 carries the Weak label. For these two countries, the robustness check therefore runs on the non-preferred specification, and the preferred Weak models, GR Real and PT Real, receive no crisis dummy test in the present exercise. This gap is listed as a clear limitation in §6.8. For Hungary and Sweden, whose preferred fallback is the nominal variant, the robustness specification and the preferred specification coincide. Table 5 reports, for each robustness tested model, the lag and rank choice, the baseline α of the house price equation, the α after including the crisis dummies and their difference $\Delta\alpha$.

Table 5: VECM Re-Estimation with GFC and COVID Crisis Dummies

Country	Spec	k_Δ	Rank	α_{HP} (with dummies)	Status
AT	Nominal	1	2	-0.0141	Estimated
AT	Real	1	2	-0.0384	Estimated

CH	Nominal	1	1	-0.0480	Estimated
CH	Real	1	1	-0.0593	Estimated
DE	Nominal	5	4	-0.0074	Estimated
DE	Real	5	3	0.0032	Estimated
ES	Nominal	4	3	-0.0195	Estimated
ES	Real	4	3	-0.0354	Estimated
FI	Nominal	2	2	-0.0515	Estimated
FI	Real	2	2	-0.0583	Estimated
GB	Nominal	4	3	-0.0303	Estimated
GB	Real	4	3	-0.0400	Estimated
GR	Nominal	1	1	-0.0003	Estimated
HU	Nominal	1	1	0.0061	Estimated
NO	Real	4	4	-0.0908	Estimated
PT	Nominal	1	2	0.0095	Estimated
SE	Nominal	1	3	-0.0324	Estimated

Note. Estimation uses identical lag order and cointegration rank as the baseline VECM (Tables D1, D2). Crisis dummies enter as exogenous regressors only; they do not alter the long-run cointegration structure. The estimated α with crisis dummies is to be compared with the baseline α_{HP} from Table E2 to assess $\Delta\alpha = \alpha_{with_dummies} - \alpha_{baseline}$.

The results show that the Solid models can be re estimated cleanly after controlling for the crisis dummies. The long run relationships therefore do not depend only on extreme periods. In most countries, α changes by only a few thousandths after the crisis dummies are included. In Austria and Switzerland, direct price adjustment becomes slightly more negative ($\Delta\alpha = -0.0091$ and -0.0059), which is plausible because crisis deviations in the baseline model are partly located in the residuals and can dampen the estimated correction speed. The Hungarian fallback path shows a clearer shift. The baseline α of $+0.0012$ increases to $+0.0061$ in the crisis dummy variant. Methodologically, this jump should be read as a signal that the high inflation and transformation dynamics of the early 1990s absorb direct price adjustment in the baseline more strongly than the crisis dummies can capture. Economically, house price adjustment remains very small in both specifications and mean reversion runs practically only through the other equations of the system. In Spain and Germany, the change relative to the baseline α is small ($|\Delta\alpha| \leq 0.001$). For Germany this means that the near-zero baseline α_{HP} is preserved under the crisis controls; for Spain, that the medium-sized baseline α_{HP} is essentially unchanged.

6 Results

6.1 Overall results

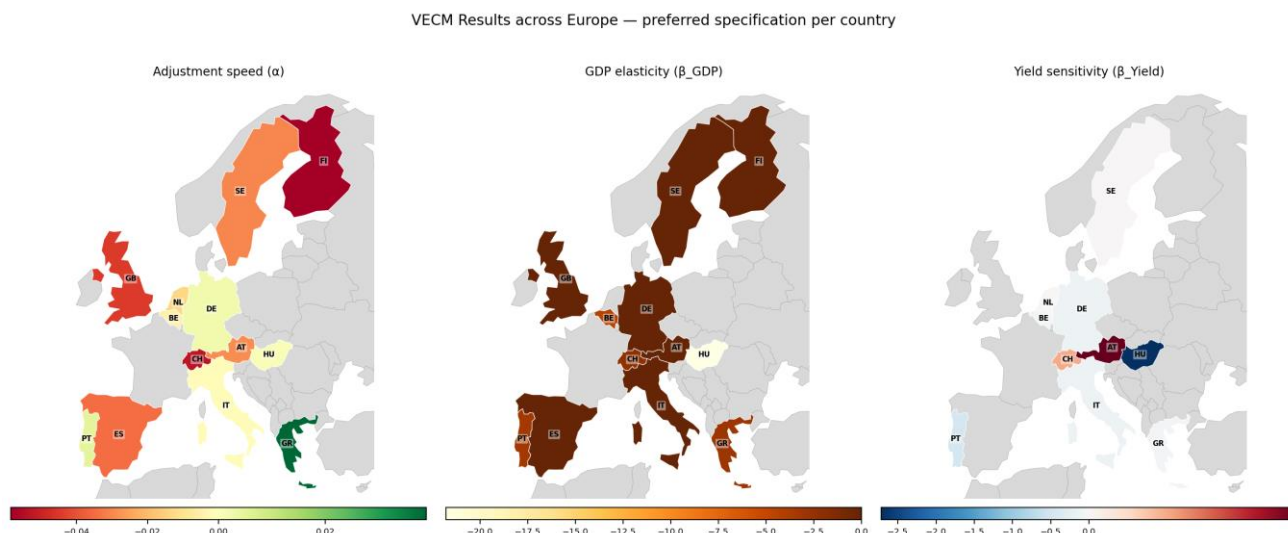
The most important overall result is that all 15 countries included in the VECM model show cointegration support. The exclusion of Denmark by the $I(1)$ filter rule is documented in §4.5 and §5.2 and not repeated here. All 15 countries included in the VECM sample show cointegration support. Given that cointegration support is a precondition for VECM admission, this count is partially induced by the sample construction. The substantive H1 evidence rests on the sign and economic plausibility of the long run β coefficients, which is examined in §6.3. This statement is stronger than a simple correlation because it refers to non-stationary time series and their joint equilibrium structure. At the same time, it must be read as a joint result across 32 Johansen tests, 16 countries times 2 specifications. At a nominal 5 percent level per test, about 1.6 false positive rank determinations would be expected under the null hypothesis. No family wise correction was applied. Country specific conclusions should therefore be read with this residual dispersion in mind (see §6.8).

H2 must be assessed in a more differentiated way. 9 preferred models are Solid, 6 are Weak and none is Not Suitable. This means that the model strategy is stable overall, but it does not reach the same diagnostic quality in every country. The evidence is methodologically strongest for Austria, Switzerland, Germany, Spain, Finland, the United Kingdom, Hungary in the nominal specification, Norway in the real specification, and Sweden in the nominal specification. For Belgium, France, Greece, Italy, the Netherlands and Portugal, the evidence remains economically interesting but diagnostically limited. Compared with the earlier nominal variant, the Solid share has fallen slightly, which is expected considering the specification decision justified in §4.5.

By fixing the real specification as justified in §4.5, we accept that the VECM diagnostics will not meet all Solid criteria for several countries. We do not interpret the higher LM share as an argument against the modelling, but as a reflection of two properties of the real series. First, the lower trend strength of deflated series compared with nominal series makes statistical identification of the cointegration relationship in the 35-year sample more difficult. Second, real series react more sensitively to structural breaks in inflation dynamics, especially the introduction of the euro, the disinflation phase from 2014 to 2021 and the inflation shock from 2022 to 2023. In most countries, the estimated β coefficients remain economically plausible in sign. We

therefore use the estimates for the misalignment series, but treat individual β coefficients with appropriate caution, see §6.8.

Figure 1: Cross-Country VECM Estimates across Europe (Preferred Specification per Country)



Note. Visual rendering of the three central VECM coefficients for the preferred per-country specification. Left panel: speed of adjustment α on the house-price equation, with diverging colour scale centred at zero — green shading marks countries where prices themselves move toward the long-run equilibrium following a deviation, red shading marks countries where adjustment occurs primarily through other system variables. Middle panel: long-run GDP elasticity β_{GDP} from the first cointegration vector. Right panel: yield sensitivity β_{Yield} . Grey-shaded countries either lie outside the panel or are excluded by the I(1) filter (Denmark). Norway is shown in grey on the β_{Yield} panel where the coefficient is normalised to zero by construction.

The consistent use of the real specification has two advantages. First, it avoids the identification distortion discussed in §4.5, where the absence of an independent CPI variable shifts the accumulated inflation component into the estimated beta coefficients of the other I(1) variables. Second, it creates a consistently real misalignment series whose deviations are not overlaid by inflation drift. This consistency is crucial when the misalignment series in §6.5 is read as a deviation from a real macroeconomic equilibrium rather than as a mix of real deviation and inflation drift. The price of this consistency is lower identification sharpness in many countries, but this is an honest limitation, not a model weakness.

For ease of reference, the sample architecture established in §5.3 is briefly recalled here. Of the original 16 countries, Denmark is excluded by the I(1) filter, leaving 15 countries in the VECM sample. Thirteen of these are modelled in the real specification, two (Hungary, Sweden) in the nominal fallback. Nine countries carry the Solid label and six the Weak label; the country-level

assignment is given in §5.3 and is the reference for the discussion in §6.3 to §6.8 and in the conclusion.

6.2 Interpretation of the figures

Appendix H assembles the diagnostic and forecast figures used throughout this paper, in three groups: in-sample fit and identification (H1–H3), economic content of the dynamics (H4–H5), and forecast assessment plus cross-country summary (H6–H10). The reading order matters because the figures are built on each other. The pure- β diagnostic in Figure H3 is the precondition for everything that follows: only if the estimated long-run relation tracks the level of the realized series can the contribution decomposition, the misalignment signal, the projection, and the coefficient comparison be read economically rather than only mechanically. All series use the preferred per-country specification reported in §5.3; only Hungary and Sweden enter on the nominal specification, the remaining thirteen on the real specification.

Figure H1 plots the realized log house-price level against the one-step-ahead VECM fit for each of the fifteen countries. The forecast uses the full short-run dynamics, namely the α -correction term plus Γ -lag dynamics, but takes the previous realized period as anchor. Visually the fit is very close in almost all panels, but this closeness is design-based: each next-period prediction reconstructs the immediately following observation from realized information up to the previous quarter, so most of the level path is carried by the anchoring rather than by the model. Figure H1 is therefore a necessary but not sufficient argument for model quality. A tight one-step fit can coexist with weak forecasts if adjustment runs primarily through the own lag rather than through the long-term cointegration relation, and this caveat must be kept in mind throughout.

Figure H2 is the differenced representation of the same one-step VECM fit. Removing the level information exposes innovation volatility and adjustment dynamics on the same scale, making residual size and sign easy to read by eye. The lag dynamics of the full VECM are still in the prediction, however, so the figure does not isolate the long-run β -implication and is not an independent identification test of the equilibrium structure. That isolation is what Figure H3 provides.

Figure H3 plots the realized log house-price level against the equilibrium level implied by the centered cointegration deviation $u_t = \beta'X_t - \text{country mean}$, with the VECM lag dynamics deliberately removed. The question is not whether the full VECM tracks the series, which H1 and H2

already establish, but whether the estimated long-run relationship by itself carries the level path. If the pure- β implication systematically matches the realized level, the equilibrium structure is economically identified, which is the precondition for H4, H5, H6/H7 and H8/H9 to be read economically.

Across the fifteen countries the quality of the match varies noticeably. In Switzerland, the United Kingdom, France, Sweden and Norway the equilibrium-implied level visibly tracks the realized path. In Belgium, Italy and the Netherlands the dashed equilibrium line is close to flat against a clearly trending actual, indicating that the estimated long-run relation captures little of the level dynamics. Hungary and Portugal show large equilibrium-implied swings that overshoot the realized series, consistent with the wider caveat on Hungary's nominal specification (§5.3, §6.8). Where the pure- β fit is weak, the per-country economic reading downstream must be downgraded accordingly.

Figure H4 is the bridge between the statistical fit and the economic interpretation. Stacked bars decompose the annual change in log house prices into contributions from the own lag (light blue, persistence), GDP per capita (dark blue, income), inflation (orange, nominal environment), the 10-year yield (yellow, capital-market channel), and the long-run adjustment plus residual (grey); the black line is the realized annual $\Delta \log$. Where price changes are dominated by grey, the dynamics are carried by the long-run correction term and unexplained residual movement, which is a warning sign for economic interpretability. H3 and H4 are needed together because H4 alone would leave open whether the contributions come from the long-run structure or from the lag dynamics.

Empirically, persistence is the largest visible component in almost every country and the second largest grey component. The 10-year yield is most visible in Norway and at smaller magnitudes in Austria, France and Portugal. The GDP-per-capita contribution is small in level terms in nearly every panel, consistent with the long-run elasticities in Figure H10, but worth flagging because it means the income channel acts on the price level chiefly through the long-run β -vector rather than through visible short-run contributions. The decomposition turns the model into a screening tool: yield-driven markets are more exposed to capital-market reversals, residual-dominated markets must be read more cautiously, and the H3 diagnosis tells us country by country whether the screening rests on identified structure or on lag mechanics.

Figure H5 turns the cointegration deviation $u_t = \beta'X_t - \text{country mean}$ into the operational misalignment series used in §6. Red shading marks overvaluation in the model's sense, blue marks undervaluation; magnitudes are in log points ($0.20 \approx 22\%$ in level terms). H5 is the direct economic counterpart to H3: where H3 establishes whether the equilibrium structure is identified, H5 turns the deviation from that equilibrium into an interpretable series. Most panels show recognizable alternating phases of over- and undervaluation, generally within ± 0.4 log points. Hungary is the exception: it uses the nominal specification on a 26-year sample and exhibits log-point swings well beyond the range of the other countries, and per the caveat in §5.3 and §6.8 its values are read qualitatively only. The same logic applies in weaker form to any country in which the H3 fit is poor.

Figures H6 and H7 are the graph and the table of the same exercise: a 12-quarter pseudo-out-of-sample backtest covering 2018Q1–2020Q4, with the cointegration structure held fixed at the values estimated through 2017Q4. H7 reports actual versus forecast cumulative growth in percent, the forecast miss in percentage points (heat-coded), and RMSE and MAE on log levels. Together they provide the genuine forecast diagnostic that H1 and H2 cannot, because the training-window cutoff means the model cannot see the realized next period when forecasting.

The test window deliberately includes the early COVID quarters, which produced large house-price moves no pre-2018 information set could anticipate. Misses below ± 3 pp on cumulative growth — Belgium, Switzerland, Finland, Hungary, the Netherlands, Norway, Sweden — are strong outcomes against that benchmark. Misses of 5–10 pp — Austria, France, Italy, Great Britain — remain useful for a 12-quarter horizon and are typical of a structural model run through a regime shift. The four outliers are Spain (+24.3 pp, over-prediction), Portugal (–23.8 pp, under-prediction), Greece (–15.3 pp, under-prediction) and Germany (+10.6 pp, over-prediction); the first three are also the three largest RMSE and MAE values. The RMSE/MAE ranking matters more than the headline cumulative miss, because the cumulative number can be small for a forecast that is consistently off in opposite directions, whereas RMSE/MAE penalizes misses period by period. The pattern is consistent with the H3 reading: countries with strong pure- β fit (Switzerland, Norway, Sweden) sit in the small-miss group, while weak pure- β fit combined with large grey residuals or extreme misalignment swings (Greece, Portugal) appears in the large-miss group.

Figures H8 and H9 are the graph and the table of the conditional VECM projection forward to end-2026, estimated on the full available sample. H8 plots historical actuals (last ten years), point forecast, and 90% pointwise confidence band; H9 ranks countries by point-estimate cumulative growth and reports the spec, latest quarter, forecast end, expected growth (heat-coded), and 90% pessimistic and optimistic bounds.

The point estimates span a wide cross-section. The clearest projected upward movements appear in Finland, Hungary and Spain; the clearest downward movements in Norway, Austria and Italy. Ten of fifteen countries have point-estimate growth at or above zero, five (the Netherlands, Sweden, Italy, Austria, Norway) below. The exact per-country figures are given in §6.6 and Figure H9. The 90% bands are wide for most countries and visibly asymmetric in some: several "negative-growth" countries have optimistic bounds reaching 5–8% appreciation, and several "positive-growth" countries have pessimistic bounds in clearly negative territory. The bands reflect cumulative innovation variance over the horizon and should not be read as narrow point forecasts. Hungary's band of -6.8% to $+32.9\%$ is real but amplified by the nominal specification and the long sample and read qualitatively. The projection is conditional on the cointegration structure remaining valid out of sample; structural breaks beyond the estimation window are not priced into the band.

Figure H10 puts the three central VECM parameters on the European map. The left panel shows the speed of adjustment α on the house-price equation on a diverging scale centered at zero: green countries are those in which prices themselves move toward the long-run equilibrium, red countries those in which adjustment occurs primarily through the other system variables. The middle and right panels show the long-run GDP elasticity β_{GDP} and the yield sensitivity β_{Yield} from the first cointegration vector. Denmark appears in grey because it is excluded by the $I(1)$ filter; Norway is grey on the β_{Yield} panel because its yield coefficient is normalized to zero by construction.

Read with H3, the α -panel tells us where long-run identification is also accompanied by the price equation actively closing the gap, as opposed to the gap being closed elsewhere. Read with H4, the β -panels indicate which fundamental variable carries the long-run elasticity in each country, and therefore which macro shock the price level is most sensitive to. This cross-sectional view summarizes the structural heterogeneity that H3, H4 and H5 already imply panel by panel, and ties it back to forecast performance in H6/H7 and the projection in H8/H9.

Figures H1 and H2 establish that the full VECM, given the previous quarter, can reproduce the next; this is necessary but cheap. Figure H3 is the decisive identification test. Figure H4 attributes short-run movement to fundamentals and to long-run adjustment; Figure H5 turns the long-run residual into the misalignment signal used in §6. Figures H6 and H7 provide the genuine out-of-sample test on a deliberately demanding window; Figures H8 and H9 carry the same identified structure forward; Figure H10 closes the loop with the cross-sectional pattern. Where the chain holds across all six steps, the VECM functions as a screening tool for over- and undervaluation. Where one or more steps weaken, the screening must be read more cautiously, with Hungary the most explicit example and Greece, Portugal, Belgium, Italy and the Netherlands as cases in which the chain weakens at specific links rather than across the board.

6.3 Hypothesis assessment

The existence component of H1 is fulfilled by sample construction for all 15 countries that pass the $I(1)$ filter and the Johansen trace test. This is therefore not the substantive content of the H1 result. The substantive content is that, in 12 of the 15 countries, the estimated long run β coefficients carry the expected signs (positive on log GDP per capita, negative on the long-term yield where present), and that the pure beta diagnostic in Figure H3 shows an economically identified level path. In Germany, Greece and Portugal, the adjustment runs through other equations of the system rather than through the house price equation itself, with $\alpha_{HP} \geq 0$. H1 is therefore confirmed in its existence and sign components for the majority of the sample, but the strength of the evidence varies country by country.

H2 is partially confirmed. The model architecture is robust enough to deliver stable preferred specifications for all 15 countries in the VECM sample. Six countries show residual autocorrelation at the Bonferroni corrected level (BE, FR, GR, IT, NL, PT) and receive the Weak label for exactly this reason. Methodologically, this is not a failure but a transparent result. The paper shows which countries are strongly reliable and which provide only qualified evidence.

H3 is examined descriptively through the pseudo out of sample test specified in §4.10 and the path-error metrics documented in §6.4. The reading proceeds along the median MAE and median RMSE between the Solid and Weak groups, with qualitative path fidelity in Figure H6 as additional context. As stated in §3 and §4.10, this examination is not framed as a hypothesis test at the present sample size; the country-level conclusions are reported in §6.4 and consolidated in §7.

6.4 Forecast performance

The pseudo out of sample test from §4.10 examines how closely the VECM paths remain to the realised log house price path in the 12 quarters after $T_c = 2017Q4$. Figure 3 reports the specification used for each country, the train and test boundaries, the realised forecast horizon H and the two path error metrics RMSE and MAE at log level.

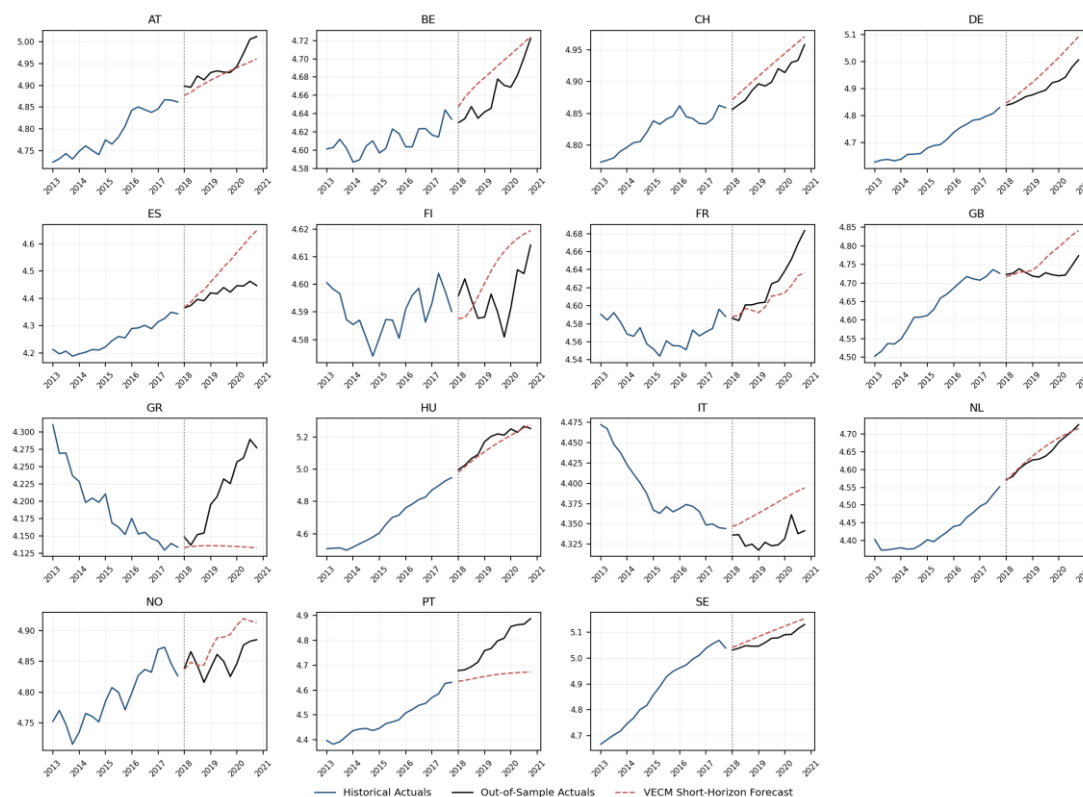
The central reading of the tables follows three questions.

First, the absolute level of the path errors is relevant. At log level, RMSE values below about 0.05 can be classified as economically small. They correspond to cumulative path deviations of less than about five percent over three years and lie within the range of typical quarterly volatility in European house price series. Values between 0.05 and 0.15 are moderate. Values above 0.15 indicate substantial divergence between the modelled and realised path. 10 of 15 countries have an RMSE below 0.05 (AT, BE, CH, FI, FR, HU, IT, NL, NO, SE). 5 countries are in the moderate range between 0.05 and 0.15 (DE, ES, GB, GR, PT), with Portugal at RMSE = 0.1373 close to the upper threshold of 0.15.

Second, the comparison between Solid and Weak countries is the descriptive object for H3. The expectation is that the median RMSE and the median MAE are lower in the Solid countries than in the Weak countries. The data deliver an asymmetric picture. In MAE, the Solid countries, with a median of 0.0287, are about 0.0029 log units below the Weak countries, whose median is 0.0316. The expected order is fulfilled. In RMSE, the order reverses narrowly. The Solid median of 0.0357 is about 0.0015 log units above the Weak median of 0.0342. This inversion is substantially driven by two Solid countries with high path errors (Spain 0.1050; Germany 0.0646) and two Weak countries with very low path errors (the Netherlands 0.0141; France 0.0215). Therefore, only one of the two path error measures delivers the order expected in §4.8, while the other shows an inversion. The median differences are small in both measures (≤ 0.003 log units) and are not statistically secured given the available sample sizes ($n_{\text{Solid}} = 9$, $n_{\text{Weak}} = 6$). The median differences in both metrics are within the range that would be expected from sampling noise alone at $n_{\text{Solid}} = 9$ and $n_{\text{Weak}} = 6$, and a formal test would have insufficient power to distinguish them from zero. The numbers therefore neither support nor reject the diagnostic-quality link; on the present test design they are silent on it.

Third, the qualitative path fidelity in Figure 2 is relevant. Individual countries can have a relatively low RMSE and still miss the movement direction of the realised path. This can occur, for example, when the VECM path draws a flat mean reversion while the realised series shows a clear trend. Such cases are diagnostically relevant because they make the mean reversion effect of the VECM visible in its purest form. Candidate cases in the present test are the Netherlands, Italy and Belgium. All three show low path errors (NL 0.0141; IT 0.0409; BE 0.0275) with very small direct house price adjustment ($\alpha_{NL} = -0.0135$, $\alpha_{IT} = -0.0013$, $\alpha_{BE} = -0.0052$). In these cases, the close point fit should be read less as forecasting strength and more as a coincidental match between the realised path and a sideways VECM mean reversion. The model meets the realised path because the path itself moves sideways in the test window, not because the model actively forecasts it. This reading is symmetric. Italy is itself classified as Weak. Within the Solid class, only Germany ($\alpha_{HP} = +0.0028$) and Hungary ($\alpha_{HP} = +0.0012$, nominal fallback) have very small $|\alpha_{HP}|$, and both already carry the diagnostic caveat from §5.4. Outside DE and HU, no Solid country has $|\alpha_{HP}|$ close enough to zero for this caution to apply, so the symmetric concern is contained to those two cases.

Figure 2: Short horizon VECM forecast versus realized path by country (backtest 2018Q1 to 2020Q4, log level)



Note. Pseudo-out-of-sample forecast paths over a 12-quarter horizon, holding the cointegration structure fixed at the values estimated up to 2017Q4. Solid blue: training-window historical actuals; solid black: out-of-sample realised log house prices; dashed red: VECM short-horizon forecast. The vertical dotted line marks the train/test cutoff. The plot illustrates the country-by-country path discussion in §6.4.

Figure 3: Backtest Accuracy — Actual vs. Forecasted Cumulative Growth (Short-Horizon VECM, 2018–2020)

Backtest Accuracy — Actual vs. Forecasted Growth (Short-Horizon VECM, 2018–2020)

Country	Period	Actual Growth %	Forecast Growth %	Forecast Miss %	RMSE	MAE
AT	2018-01-01 – 2020-10-01	12.0	6.4	-5.6	0.0257	0.0201
BE	2018-01-01 – 2020-10-01	9.7	9.9	0.2	0.0275	0.025
CH	2018-01-01 – 2020-10-01	10.7	12.1	1.4	0.0209	0.0199
DE	2018-01-01 – 2020-10-01	18.3	28.9	10.6	0.0646	0.0575
ES	2018-01-01 – 2020-10-01	8.6	32.9	24.3	0.105	0.0841
FI	2018-01-01 – 2020-10-01	1.9	2.4	0.5	0.0151	0.0131
FR	2018-01-01 – 2020-10-01	10.2	5.2	-5.0	0.0215	0.0165
GB	2018-01-01 – 2020-10-01	5.1	12.5	7.4	0.0519	0.0406
GR	2018-01-01 – 2020-10-01	13.8	-1.5	-15.3	0.0928	0.0767
HU	2018-01-01 – 2020-10-01	29.2	32.4	3.1	0.0357	0.0287
IT	2018-01-01 – 2020-10-01	0.5	6.0	5.4	0.0409	0.0382
NL	2018-01-01 – 2020-10-01	16.8	15.5	-1.3	0.0141	0.0111
NO	2018-01-01 – 2020-10-01	4.8	7.7	2.9	0.0372	0.0316
PT	2018-01-01 – 2020-10-01	23.2	-0.6	-23.8	0.1373	0.1228
SE	2018-01-01 – 2020-10-01	10.4	12.9	2.5	0.0288	0.0271

Note. Tabulated rendering of the short-horizon backtest with country-level actual versus forecasted cumulative growth (in percent), forecast miss (forecast minus actual, in percentage points), RMSE and MAE on log levels. The "Forecast Miss %" column is heat-coded: green for over-prediction, red for under-prediction; magnitude scales with colour saturation.

6.5 Misalignment at the latest observation

The misalignment series m_t^{HP} defined in §4.9 turns the estimated cointegration structure into an interpretable level signal. Whereas Figure H5 traces the full historical path of u_t per country, the operationally relevant value for a screening tool is the most recent observation. Table 6 reports m_t^{HP} at the latest available quarter per country, expressed both in log points and in percent deviation ($\exp(m_t^{\text{HP}}) - 1$). Positive values indicate observed prices above the long-run level explained by the country-specific fundamentals; negative values indicate the opposite. The diagnostic label from §4.8 is repeated to remind the reader that misalignment values from Weak countries should be read with the caveats discussed in §6.2 and §6.8.

Table 6: Misalignment at the Latest Available Quarter, by Country

Country	Spec	Latest quarter	m_t^{HP} (log points)	Deviation (%)	Diagnostic label
AT	Real	2025Q1	0.239	27.0	Solid
BE	Real	2025Q1	-0.055	-5.3	Weak

CH	Real	2025Q1	-0.041	-4.0	Solid
DE	Real	2025Q1	0.202	22.3	Solid
ES	Real	2025Q1	0.208	23.2	Solid
FI	Real	2025Q1	-0.023	-2.2	Solid
FR	Real	2025Q1	0.056	5.7	Weak
GB	Real	2025Q1	0.289	33.6	Solid
GR	Real	2025Q1	0.075	7.7	Weak
HU	Nominal	2025Q1	-7.375	n/a	Solid (caveat)
IT	Real	2025Q1	-0.250	-22.1	Weak
NL	Real	2025Q2	0.422	52.5	Weak
NO	Real	2025Q1	0.487	62.8	Solid
PT	Real	2025Q1	0.622	86.2	Weak
SE	Nominal	2025Q1	0.763	114.5	Solid

Note. m_t^{HP} is the projection of the centred cointegration deviation u_t onto the house-price equation, defined in §4.9. Positive values indicate observed prices above the long-run level implied by the country-specific fundamentals; negative values indicate the opposite. The percent deviation is computed as $(\exp(m_t^{\text{HP}}) - 1) \times 100$; for Hungary this transformation is suppressed and reported as n/a, because the nominal fallback specification on a 26-year sample yields a numerically fragile β -vector whose exponentiated reading is economically uninformative. The Hungarian log-point value is retained for transparency only and is read qualitatively (see §5.3 and §6.8). The values in this table are not absolute valuation levels. By construction (§4.9) the misalignment series is mean-zero across the country-specific estimation window. The entries therefore measure how far the latest observed price lies above or below the country's own historical relationship between price and fundamentals, not above or below an external normative reference. A persistent secular shift in unobserved fundamentals over the estimation window — for instance, a long-run decline in equilibrium real interest rates not captured by the modelled yield series — would be absorbed into the centred reference and would not appear in this table as a positive misalignment.

The table reveals a striking quantitative pattern. The largest positive misalignments, in descending order, appear in Sweden (+114.5 %, nominal fallback), Portugal (+86.2 %, Weak), Norway (+62.8 %, Solid), the Netherlands (+52.5 %, Weak), the United Kingdom (+33.6 %, Solid), Austria (+27.0 %, Solid), Spain (+23.2 %, Solid), and Germany (+22.3 %, Solid). Italy (-22.1 %, Weak) is the only country with a substantial undervaluation in the real specification; Belgium, Finland and Switzerland show small deviations near zero. The Hungarian value is reported with a caveat in the table itself because it reflects the numerical boundary of the nominal β -vector identified on the 26-year sample (§5.3) and is not used for quantitative cross-country comparisons.

Three readings of this pattern are appropriate. First, the dominant signal across the European panel is overvaluation rather than undervaluation: of the fourteen interpretable countries, ten

are above their estimated long-run level, and the magnitude of the deviations is larger in absolute terms than the corresponding undervaluations. Second, the misalignment sign is not mechanically linked to the diagnostic label: Solid countries appear with both deviations of close to zero (Switzerland, Finland) and very large overvaluations (Norway, the United Kingdom, Sweden). The diagnostic label refers to the reliability of the estimated equilibrium, not to the direction of the deviation from it. Third, the operational reading of these values must respect the pure- β fit in Figure H3. Countries with strong level-tracking in Figure H3 — Switzerland, the United Kingdom, France, Sweden, Norway — yield misalignment values that the screening logic is designed to produce. In countries with weak pure- β fit (Belgium, Italy, the Netherlands), the m_t^{HP} value is technically defined but should be read with the level-tracking caveat from §6.2. Sweden's large positive value additionally reflects the nominal fallback specification and is not directly comparable to the real-specification countries.

The link between these misalignment values and the conditional forward path through 2026Q4 is taken up in §6.7.

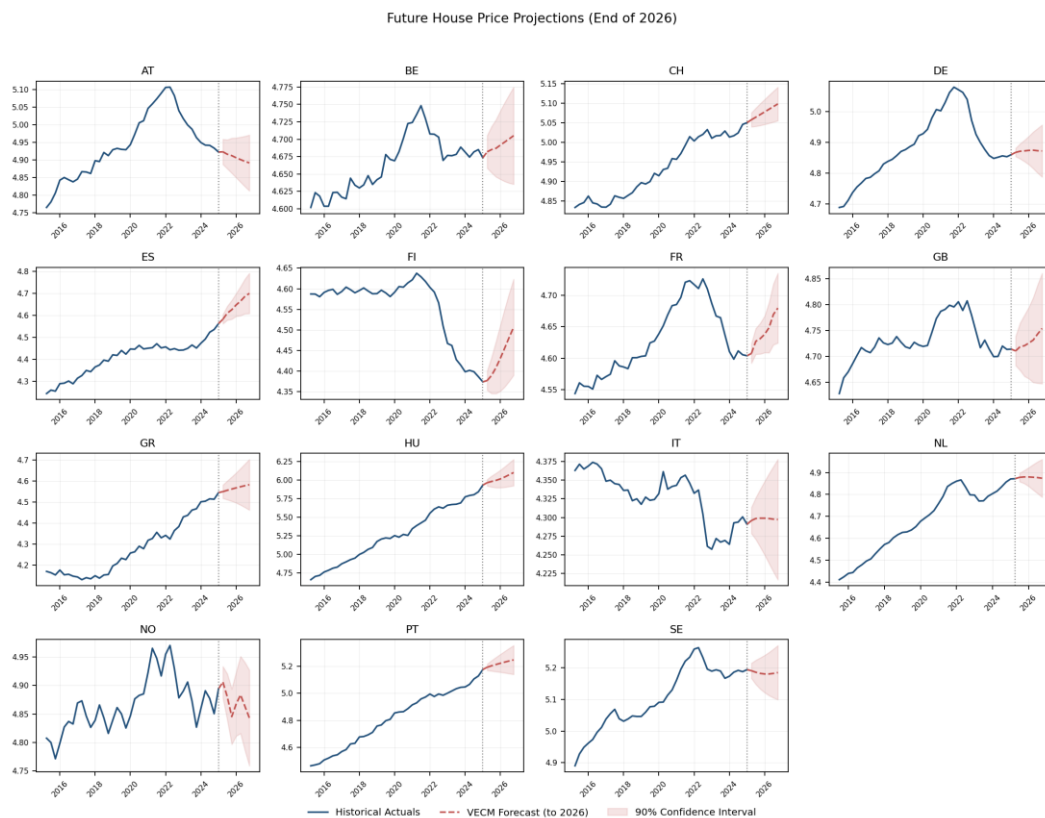
6.6 Forward-looking Projection

The forward-looking projection specified in §4.11 uses the full available data history for each country and calculates the conditional log house price path to the end of 2026, each with a pointwise 90 percent confidence band.

The economic reading proceeds in three steps. First, the sign of $\Delta \log \text{HP}$ is relevant. Countries with a positive sign follow an upward path in the conditional forecast. Countries with a negative sign follow a downward correction. Countries close to zero follow a sideways movement. Counting across the 15 countries yields 9 countries with an upward path ($\Delta > +0.01$: BE, CH, ES, FI, FR, GB, GR, HU, PT), 4 countries with a sideways path ($|\Delta| \leq 0.01$: DE, IT, NL, SE) and 2 countries with a downward path ($\Delta < -0.01$: AT, NO). The clearest projected upward movements appear in Hungary (+0.1365), Finland (+0.1298) and Spain (+0.1173). The clearest downward movement appears in Norway (−0.0628). Second, CI width is a complement to diagnostics. Countries with Solid diagnostics and a narrow CI are those in which the projection has the highest explanatory value. Switzerland stands out clearly here with a CI width of 0.0861. Countries with Weak diagnostics and wide CIs, Greece (0.2420) and Portugal (0.2134), should not be read as robust point forecasts, but as indications of a broadly dispersed band that cannot be tightly pinned down. Hungary's CI width of 0.3551, measured as the difference between upper and lower

bounds at log level, reflects the high inflation history of the sample in the nominal fallback estimate and is no longer informative as a point forecast. It corresponds to a multiplicative 90 percent range of about 1.43x between lower and upper bound ($\exp(0.3551) \approx 1.43$). Expressed as a symmetric band around the point forecast, this means about 1.19x in each direction ($\exp(0.3551/2) \approx 1.19$). The Hungarian projection should therefore be read only qualitatively as an upward tendency. A quantitative use of the endpoint is methodologically not robust. Third, the projection is always subject to the condition that the equilibrium structure identified in the full sample continues to hold. In particular, the path assumes a continuation of the inflation and interest rate regime visible in the sample. A clear future structural break, whether a renewed inflation shock or a sudden interest rate reversal, would shift the path in a way that is only incompletely captured by the 90 percent bands.

Figure 4: Future House Price Projections to End-2026 (Conditional VECM Path, 90% CI)



Note. Forward-looking VECM projection of the log house-price level per country, estimated on the full available data sample. Solid blue: historical actuals (last ten years shown); dashed red: conditional point forecast to 2026Q4; red shading: 90% pointwise confidence band. The vertical dotted line marks the start of the projection window. Country-specific spec choice (Real or Nominal) follows §5.3 and is reported alongside Figure H9.

Figure 5: Future House Price Projections to 2026 — Ranked by Expected Growth (90% CI)

Future House Price Projections to 2026 — ranked by expected growth (VECM, 90 % CI)

Country	Spec	Latest Data	Forecast End	Expected Growth %	Pessimistic (90 CI)%	Optimistic (90 CI)%
FI	Real	2025-07-01	2026-10-01	15.5	2.8	29.8
HU	Nominal	2025-07-01	2026-10-01	11.3	-6.8	32.9
ES	Real	2025-07-01	2026-10-01	8.7	-0.8	19.1
FR	Real	2025-07-01	2026-10-01	7.3	1.6	13.4
GB	Real	2025-10-01	2026-10-01	5.0	-5.6	16.9
GR	Real	2025-07-01	2026-10-01	1.9	-9.7	15.0
CH	Real	2025-10-01	2026-10-01	1.5	-2.8	5.9
BE	Real	2025-07-01	2026-10-01	0.7	-6.1	7.9
DE	Real	2025-07-01	2026-10-01	0.6	-7.6	9.4
PT	Real	2025-07-01	2026-10-01	0.3	-9.8	11.6
NL	Real	2025-07-01	2026-10-01	-0.5	-8.7	8.4
SE	Nominal	2025-10-01	2026-10-01	-1.5	-9.5	7.4
IT	Real	2025-07-01	2026-10-01	-1.8	-9.4	6.4
AT	Real	2025-07-01	2026-10-01	-2.6	-10.0	5.5
NO	Real	2025-10-01	2026-10-01	-3.4	-11.2	5.1

Note. Tabulated rendering of the forward-looking projection in Figure H8, ranked by expected cumulative real (or nominal, where flagged) growth from the latest data point to 2026Q4. The "Expected Growth %" column is heat-coded: green for positive expected growth, red for negative, with saturation proportional to magnitude. Pessimistic and optimistic columns report the 90% confidence interval bounds.

6.7 Misalignment and projection: a cross-check

Two outputs of the country-level VECM speak to opposite ends of the price cycle. The misalignment value at the latest observation (§6.5) measures how far the observed log house price currently lies above or below the long-run level implied by the estimated cointegration structure. The conditional forward projection (§6.6) describes the path that the same model expects through 2026Q4, holding the cointegration structure fixed. The two outputs are related but not identical: the projection contains the misalignment-driven pull ($\alpha_{HP} \times m_t^{HP}$), the Γ -lag short-run dynamics, and the endogenous evolution of GDP, inflation and yields within the system. A direct one-to-one mapping between misalignment sign and projection sign is therefore not expected.

The cross-check assesses three patterns. The first pattern is clean directional match, where the misalignment sign and the projection sign agree under a substantially negative α_{HP} . Norway is the cleanest case: a real-specification overvaluation of +62.8 % combines with $\alpha_{HP} = -0.0886$

(the strongest direct adjustment in the panel) and a downward projection of -6.3% to 2026Q4. Austria shows the same logic at smaller magnitudes (overvaluation $+27.0\%$, downward projection). Switzerland matches in the opposite direction: a small undervaluation of -4.0% with $\alpha_{HP} = -0.0534$ and an upward projection. Belgium fits with caveats because both magnitudes are small and the model is Weak.

The second pattern is decoupling, where the projection does not respond mechanically to the misalignment because the price equation itself does not contain the adjustment channel. Germany and Italy have α_{HP} near zero, so the misalignment value ($+22.3\%$ and -22.1% respectively) does not translate into a price-side correction within the forecast horizon; both show sideways projections, with the adjustment running through the GDP, inflation and yield equations. The Netherlands shares the same logic: a small $|\alpha_{HP}|$ of 0.0135 , with the misalignment of $+52.5\%$ running through other equations rather than triggering a price-side correction over the projection horizon. Greece and Portugal carry positive α_{HP} estimates, which means the price equation is not the equilibrating one and the projection is dominated by short-run dynamics rather than by mean reversion.

The third pattern is internal model inconsistency, where the misalignment is large and α_{HP} is meaningfully negative, yet the projection runs against the implied error-correction direction. Spain ($+23.2\%$, $\alpha_{HP} = -0.0348$, projection $+11.7\%$), the United Kingdom ($+33.6\%$, $\alpha_{HP} = -0.0445$, projection upward) and Sweden ($+114.5\%$, $\alpha_{HP} = -0.0300$, projection sideways) fall into this category. The implied 12-quarter α -pull in these countries would be in the range of -0.10 to -0.34 log points, which is not negligible. The fact that the projection nevertheless runs upward or sideways means that the Γ -lag short-run dynamics and the endogenous macro path within the VECM dominate the cointegration-driven correction over the projection horizon. Two readings of this tension are possible. The economic reading is that the equilibrium structure suggests these markets are above their long-run level, but the model does not see the conditions for a correction within the projection window; the correction, if it materialises, would lie beyond it. The methodological reading is that when the short-run dynamics systematically override the α -correction over three years, the operational weight that should be placed on the misalignment value as a screening signal in these countries is correspondingly reduced. We do not adjudicate between the two readings here. Both lower the confidence with which the misalignment values

for ES, GB and SE can be used as a forward-looking signal, and the screening conclusions in §2.4 and §7 are read accordingly.

This three-pattern reading replaces a simple consistency check with a more honest classification. It also sharpens the screening logic introduced in §2.4. The cleanest screening signals come from the first pattern (Norway, Austria, Switzerland), where misalignment, adjustment speed and projection point in the same direction. The third pattern (Spain, the United Kingdom, Sweden) carries the lowest operational confidence in the present cross-check, because the equilibrium-based overvaluation signal and the model's own projection deliver opposite messages and cannot both be acted on. The Hungarian fallback case (§5.3) is excluded from this cross-check because the underlying β -vector is identified at the numerical boundary; both the misalignment value and the projection rest on the same fragile estimate and cannot be cross-checked against each other. A formal test, for example a cross-country regression of cumulative projected return on the latest misalignment value, possibly weighted by α_{HP} , is feasible with the present output and is identified as a follow-up step.

6.8 Limitations

This chapter consolidates the methodological boundaries of the analysis. The boundaries are listed in descending order of importance for the interpretation of the results.

A first boundary lies at the macroeconomic scope of the model. The VECM works with macroeconomic quarterly data. Asset quality, rent regulation, new construction volume, credit standards, taxes, vacancy, local supply restrictions and institutional investor demand are not modelled directly. The model is therefore designed for country-level screening and is not intended to support the valuation of an individual asset.

A second boundary concerns variable selection. Income, inflation and yields are central fundamentals, but they do not capture every relevant mechanism. The originally intended population variable was not retained in any country because the ADF-based $I(1)$ filtering does not classify the logged population series as $I(1)$ under the chosen test specification (see §4.6). The demographic demand component is therefore not represented directly in the system. Credit standards, debt, new construction supply, rent regulation, tax incentives, land policy and international capital flows are likewise outside the core model. Their omission is partly data-driven and partly deliberate, because an overloaded model across 16 countries would reduce

comparability. These factors can nevertheless play a relevant role in individual countries. A related point concerns the integration-order treatment of the population variable. The ADF tests applied here use a constant-only specification. A Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test or an ADF test with deterministic trend could in principle yield a trend-stationary classification for log Pop in some countries, in which case the variable could re-enter the system in differences with a deterministic trend term. We do not pursue this alternative here because the $I(1)$ filter rule was set ex ante for the entire candidate set, and a country-specific deviation for population alone would compromise the consistency of the variable selection across countries. The chosen filter treats two econometrically distinct cases, namely $I(2)$ integration and trend-stationary persistence, as observationally equivalent within the present setup. Either reading is consistent with the exclusion of log Pop from the $I(1)$ system, although on closer inspection they would motivate different model architectures (an $I(2)$ -consistent specification or a level VAR with deterministic trend). A targeted re-examination of the demographic channel under these alternative specifications is identified as a useful follow-up step.

A third boundary concerns the interpretation of several cointegration vectors. At high rank, the presentation of individual beta coefficients depends on identification and normalisation. For this reason, the paper focuses on dominant drivers, β -values, diagnostics and figures. The coefficients are not read as exact structural parameters. This caution is methodologically appropriate and supports the credibility of the results.

A fourth boundary concerns the specification choice itself. The decision in §4.5 to use the real house price series is econometrically and economically consistent. It comes with the trade-off that the cointegration relationship in the 35-year sample is less sharply identified in some countries than in the nominal variant. In the present real specification, six countries are classified as Weak. A comparable consideration applies to the fallback countries Hungary and Sweden, whose estimation relies on the nominal variant. Their misalignment values are not directly comparable to the real misalignment values of the other countries and are therefore read with appropriate caution in §6.5 and §6.7.

A fifth boundary concerns the design of the out-of-sample test. The path-error analysis compares the VECM forecasts with realised values, but not with alternative time-series benchmarks such as a country-specific $AR(1)$ or a random walk. A formal Diebold-Mariano test of equal predictive accuracy by country and horizon is therefore not part of this exercise. Consequently, H3

is examined in the form of whether the estimated equilibrium structure traces the realised path in economically plausible proximity and whether the diagnostic labels carry separating power for path quality. It is not examined in the sharper form of whether the VECM information set extends beyond what a pure persistence model would deliver. We acknowledge this restriction transparently. It narrows the reach of the H3 claim while leaving its substantive content intact, because the contribution of the VECM in this paper lies in the macroeconomic identification of the equilibrium structure and the misalignment series, not in a comparative exercise against atheoretical time-series benchmarks. The benchmark comparison and the formal Diebold-Mariano test are identified as the most important methodological extension for follow-up work.

A sixth boundary concerns the heterogeneity in training-sample length across countries. The effective training length varies between 19 and 28 years (see §4.10). For countries with shorter training samples (HU 19, GR 21, IT 22, AT/BE/CH/ES/PT 23 years), the estimation uncertainty of the long-run coefficients is higher than for countries with full coverage from 1990. The estimated cointegration relationship in these countries is identified on a smaller information set, which mechanically widens the dispersion of the path errors in the backtest and the confidence bands in the forward projection. This effect is reflected in the country-level discussion in §6.4 and §6.6.

A seventh boundary concerns the design of the test window itself. The fixed test window 2018Q1 to 2020Q4 includes the COVID shock 2020Q1 to 2020Q2 and ends before the inflation shock of 2022/23. This asymmetry follows from the fixed 12-quarter horizon starting from 2017Q4 and is not the result of a differentiated shock selection. It remains an open empirical question whether the estimated equilibrium structure would carry the inflation structural break of 2022/23. A more comprehensive validation design would compare several training cutoffs and therefore several test windows against each other. A consequence of the present design is that the out-of-sample exercise does not extend across the most recent macroeconomic episode. A systematic validation with shifted training cutoffs is identified as a follow-up step.

An eighth boundary concerns the scope of the crisis-dummy robustness check. The check was diagnostically applied to the Solid models from §4.8. For Greece and Portugal, whose preferred specification according to §5.3 is the real Weak variant, the robustness test therefore runs on the nominal Solid variant rather than on the preferred specification. The preferred Weak models GR Real and PT Real are not subjected to the crisis-dummy test in the present exercise. The other four Weak countries (BE, FR, IT, NL) lie outside the robustness check because neither of

their two specifications is Solid. The H2 statement on stability under crisis control therefore covers nine countries directly. A consistent robustness check across all preferred specifications, including those classified as Weak, is identified as a follow-up step. The fallback rule itself is unchanged: the preferred-specification selection algorithm chooses the real variant as long as a VECM-eligible model (cointegration rank ≥ 1 , no estimation failure, eigenvalue modulus ≤ 1.01) exists, irrespective of its Solid/Weak status. The fallback to the nominal variant for Hungary and Sweden is triggered exclusively by the absence of a VECM-eligible real model.

A ninth boundary concerns the forward-looking projection in §6.6. Its explanatory power is conditional. It projects the path that results if the equilibrium structure estimated in the full sample continues to hold. Structural breaks beyond the sample are reflected only to the extent that their effects can be derived mechanically from the estimated reaction functions of the system. The reported 90 percent confidence bands are asymptotically correct under the assumption of stable parameters and are read as model-side dispersion rather than as risk bands against regime shifts.

A tenth boundary concerns the absence of a family-wise correction across the 32 Johansen trace tests, namely 16 countries times two specifications. At a nominal 5 percent level per test, the expected number of falsely identified cointegration ranks under the joint null is approximately 1.6. A Bonferroni correction at the family level would lower the per-test threshold to $0.05/32$, which is approximately 0.0016, and would in principle reduce the rank classification of marginal countries. We do not apply this correction for two reasons. First, the cointegration ranks are not used as the final inference object. They are an intermediate filter before the VECM step, and the economic identification of each country-specific equilibrium relationship is examined independently through the pure- β diagnostics in Figure H3 and the contribution decomposition in Figure H4. Second, a family-wise correction across heterogeneous national systems treats statistically independent country tests as one joint experiment, which would over-correct in a research design where every country represents a substantive case in its own right. We therefore retain the per-country 5 percent threshold and read country-specific rank determinations with the residual dispersion in mind. This is a deliberate methodological choice.

7 Conclusion

The paper shows that European residential property prices in the 15 countries included in the VECM model are linked to macroeconomic fundamentals in the long run. This makes fundamental misvaluation empirically analysable. From the original 16 country pool, Denmark was excluded by the $I(1)$ filter rule. Within the remaining 15-country sample, the strongest evidence comes from the nine Solid countries; the six Weak countries (BE, FR, GR, IT, NL, PT) remain relevant but must be interpreted more cautiously. The country-level Solid/Weak assignment is documented in §5.3 and §6.1. The country specific conclusions must also be read as a joint result across 32 Johansen tests without family wise correction (see §6.8). In addition, the estimation shows a full rank constellation in six countries (ES, FI, GB, NL, NO, SE), which limits the economic interpretation of the β level coefficients in these countries and shifts the reading of the misalignment series primarily to the adjustment dynamics α (see §6.8).

Methodologically, the most important contribution is the clean separation between long run equilibrium, short run dynamics and model quality. The paper avoids the simple claim that high prices automatically mean overvaluation. Instead, it asks whether prices can be explained in relation to income, inflation and capital market interest rates. This perspective is intended as a country comparable macroeconomic screening tool. It complements rather than replaces market level information such as transaction prices, rents, vacancy rates and yields, and it is explicitly not designed to support individual investment decisions.

Overall, the research hypotheses receive heterogeneous support. H1 is empirically supported in its sign and identification components for the majority of the sample, H2 is partially confirmed with six countries failing the Bonferroni-corrected residual diagnostics, and H3 is descriptively inconclusive given the sample size and the absence of a benchmark comparison.

H3 is assessed in the form specified by §4.10 and is descriptively inconclusive at the present sample size and test design. In the path error test over 12 quarters after the cutoff 2017Q4, the VECM forecasts of log house prices remain in economically plausible proximity to the realised path: the median RMSE across all 15 countries is about 0.0357 at log level and the median MAE is about 0.0287. The MAE comparison shows the expected order, with Solid countries 0.0029 log units below Weak countries; the RMSE comparison shows an inversion of about 0.0015 log units, driven by ES and DE on the Solid side and NL and FR on the Weak side. With $n_{\text{Solid}} = 9$ and $n_{\text{Weak}} = 6$ and median differences of this magnitude, a formal test would have negligible

power. We therefore do not claim a confirmed link between the in-sample diagnostic label and out-of-sample path quality; the present test is silent on this link. A sharper forecasting statement, for example that the VECM systematically beats an AR(1) or random walk benchmark in RMSE or MAE, is not made here and remains open as a follow-up step (§6.8). The backtest is additionally conducted with full-sample hyperparameters, which biases path errors downward relative to a fully recalibrated rolling-window design and reinforces the cautious reading of H3.

The paper also shows which extensions are obvious. First, benchmark models such as AR(1) and random walk should be included formally. Second, demography should be treated more cleanly econometrically, for example through an I(2) consistent specification or through more robust demographic series that avoid the level filter problem of World Bank population. Third, rent data, credit growth and supply indicators could be added. Fourth, a scenario analysis for interest rate and inflation shocks would be useful. These extensions would broaden the model family without replacing the core contribution of the present paper.

The core contribution remains the clear methodological order. Residential property prices are not evaluated in isolation, but relative to a country specific macroeconomic equilibrium. The paper shows where this relation is strong, where it remains diagnostically weaker, and how the signals in the backtest and forward projection should be read.

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Appendix A

Data Documentation

This appendix documents the temporal coverage of the underlying quarterly panel and reports descriptive statistics for each macroeconomic variable used in the empirical analysis. Sample windows differ across countries because the BIS Residential Property Price coverage and the FRED long-term yield series begin at different dates. Missing-share values reflect the fraction of empty quarters within each country's overall data window after harmonisation to quarter-start frequency. Population is quarter-interpolated from annual World Bank data; all other series are reported at native quarterly frequency or aggregated as the within-quarter mean (long-term yields).

Table A1

Country Coverage – Property Prices, GDP per Capita, and Population

Country	HP (real)	HP (nominal)	GDP per capita	Population	N
AT	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
BE	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
CH	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
DE	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1991Q1–2025Q1 (5.5%)	1990Q1–2025Q1 (2.8%)	145
DK	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
ES	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
FI	1990Q1–2025Q3 (1.4%)	1990Q1–2025Q3 (1.4%)	1990Q1–2025Q1 (2.8%)	1990Q1–2025Q1 (2.8%)	145
FR	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q1 (2.8%)	1990Q1–2025Q1 (2.8%)	145
GB	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q1 (2.8%)	1990Q1–2025Q1 (2.8%)	145
GR	1997Q1–2025Q4 (20.0%)	1997Q1–2025Q4 (20.0%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
HU	1992Q1–2025Q4 (6.2%)	1990Q1–2025Q4 (0.7%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
IT	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1996Q1–2025Q1 (19.3%)	1990Q1–2025Q1 (2.8%)	145
NL	1990Q1–2025Q3 (1.4%)	1990Q1–2025Q3 (1.4%)	1996Q1–2025Q1 (19.3%)	1990Q1–2025Q1 (2.8%)	145

NO	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q1 (2.8%)	1990Q1–2025Q1 (2.8%)	145
PT	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1995Q1–2025Q1 (16.6%)	1990Q1–2025Q1 (2.8%)	145
SE	1990Q1–2025Q4 (0.7%)	1990Q1–2025Q4 (0.7%)	1993Q1–2025Q1 (11.0%)	1990Q1–2025Q1 (2.8%)	145

Note. Each cell reports the first–last available quarter and, in parentheses, the fraction of missing quarters within the country window. N is the total number of quarter rows after harmonisation. HP = house price index (BIS RPPI); GDP per capita is real GDP divided by interpolated population; population is from World Bank annual data, quarter-interpolated geometrically.

Table A2

Country Coverage – Inflation and Long-Term Yield

Country	Inflation (CPI YoY)	10Y government yield	N
AT	1991Q1–2025Q1 (5.5%)	1990Q1–2026Q1 (0.0%)	145
BE	1991Q1–2025Q2 (4.8%)	1990Q1–2026Q1 (0.0%)	145
CH	1991Q1–2025Q2 (4.8%)	1990Q1–2026Q1 (0.0%)	145
DE	1991Q1–2025Q1 (5.5%)	1990Q1–2026Q1 (0.0%)	145
DK	1991Q1–2025Q2 (4.8%)	1990Q1–2026Q1 (0.0%)	145
ES	1991Q1–2025Q1 (5.5%)	1990Q1–2026Q1 (0.0%)	145
FI	1991Q1–2025Q1 (5.5%)	1990Q1–2026Q1 (0.0%)	145
FR	1991Q1–2025Q1 (5.5%)	1990Q1–2026Q1 (0.0%)	145
GB	1991Q1–2025Q1 (5.5%)	1990Q1–2026Q1 (0.0%)	145
GR	1991Q1–2025Q2 (4.8%)	1997Q2–2026Q1 (20.0%)	145
HU	1991Q1–2025Q2 (4.8%)	1999Q1–2026Q1 (24.8%)	145
IT	1991Q1–2025Q1 (5.5%)	1991Q1–2026Q1 (2.8%)	145
NL	1991Q1–2025Q2 (4.8%)	1990Q1–2026Q1 (0.0%)	145
NO	1991Q1–2025Q2 (4.8%)	1990Q1–2026Q1 (0.0%)	145
PT	1991Q1–2025Q2 (4.8%)	1993Q3–2026Q1 (9.7%)	145
SE	1991Q1–2025Q1 (5.5%)	1990Q1–2026Q1 (0.0%)	145

Note. Inflation is the year-over-year percentage change in the national CPI (FRED). Long-term yield is the FRED IRLTLT01 series (monthly), aggregated to quarters via within-quarter mean.

Table A3

Descriptive Statistics – Real Residential Property Price Index

Country	N	Mean	SD	Min	Median	Max
AT	144	113.0517	20.7869	79.7580	109.6938	165.2386
BE	144	85.4871	21.4124	49.4545	99.0102	115.3707

CH	144	105.7548	26.3920	75.3867	95.9388	161.3827
DE	144	119.6496	14.8972	99.1342	119.6873	160.5950
DK	144	91.9918	27.1326	45.1801	96.1403	136.8435
ES	144	77.2085	18.9255	51.3844	75.6079	118.6858
FI	143	83.1011	17.1473	49.0249	87.0771	103.3056
FR	144	83.4805	21.7203	49.5867	95.9660	112.8278
GB	144	87.8670	27.0762	43.9222	96.5160	122.4408
GR	116	84.8994	17.3958	62.1274	81.1892	119.5312
HU	136	114.4576	34.2573	63.8528	107.4982	216.1053
IT	144	83.5579	10.1865	68.5049	78.6635	105.0489
NL	143	88.3682	24.9256	42.1424	94.7414	131.5963
NO	144	89.3707	35.4680	33.3185	95.4624	144.0209
PT	144	113.7195	21.7904	80.0374	111.4231	196.1766
SE	144	91.1574	39.6312	39.1717	91.5026	163.9677

Note. Index, 2010 = 100.

Table A4

Descriptive Statistics – Nominal Residential Property Price Index

Country	N	Mean	SD	Min	Median	Max
AT	144	115.6534	46.4219	51.8562	90.0806	218.6604
BE	144	87.7236	37.9221	32.6708	96.1872	160.5939
CH	144	101.7057	31.1104	69.5435	91.7322	169.9662
DE	144	117.8604	29.9890	84.6675	101.8411	196.1000
DK	144	91.5112	40.4754	31.5890	97.8488	172.7523
ES	144	73.8820	29.8571	29.2205	77.2598	139.3137
FI	143	83.1949	26.8679	38.1270	92.2258	120.8733
FR	144	83.4048	31.9299	39.9675	97.2188	134.6029
GB	144	91.8506	44.4815	31.7681	97.9219	172.7348
GR	116	79.9129	19.8517	40.1558	77.7655	118.3852
HU	144	110.5032	91.5252	13.7611	94.9414	416.8668
IT	144	78.6467	18.1102	37.4327	84.3777	102.2252
NL	143	90.4897	41.4319	27.4848	91.3584	195.1030
NO	144	94.9310	55.1902	23.2756	90.9447	201.8960
PT	144	107.5516	41.6221	49.2198	99.4854	261.0065

SE	144	95.1211	53.4304	31.9441	89.4983	193.1446
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Note. Index, 2010 = 100.

Table A5

Descriptive Statistics – Real GDP per Capita

Country	N	Mean	SD	Min	Median	Max
AT	121	0.0086	0.0008	0.0068	0.0089	0.0097
BE	121	0.0082	0.0008	0.0065	0.0083	0.0094
CH	121	0.0199	0.0018	0.0168	0.0203	0.0231
DE	137	0.0079	0.0010	0.0064	0.0079	0.0093
DK	121	0.0835	0.0071	0.0687	0.0829	0.0984
ES	121	0.0057	0.0005	0.0045	0.0058	0.0067
FI	141	0.0080	0.0012	0.0056	0.0086	0.0094
FR	141	0.0074	0.0008	0.0060	0.0076	0.0086
GB	141	0.0086	0.0012	0.0065	0.0090	0.0102
GR	121	0.0046	0.0005	0.0037	0.0044	0.0057
HU	121	0.7208	0.1627	0.4663	0.7009	1.0096
IT	117	0.0068	0.0003	0.0056	0.0068	0.0073
NL	117	0.0096	0.0009	0.0074	0.0097	0.0112
NO	141	0.1292	0.0176	0.0902	0.1356	0.1539
PT	121	0.0042	0.0003	0.0034	0.0042	0.0049
SE	129	0.0906	0.0127	0.0637	0.0950	0.1075

Note. Real GDP per capita as Real_GDP / interpolated population. Levels reflect the source-currency chain-linked GDP series divided by population, so cross-country comparisons of absolute levels are not directly meaningful; only the within-country trend is used in the VECM.

Table A6

Descriptive Statistics – CPI Inflation (YoY, %)

Country	N	Mean	SD	Min	Median	Max
AT	137	2.4402	1.7784	0.0310	1.9855	10.5950
BE	138	2.2854	1.7433	-1.2189	2.0848	11.0773
CH	138	0.9786	1.4416	-1.3741	0.6555	6.2500
DE	137	2.0722	1.6486	-0.9165	1.6429	8.5805
DK	138	1.9502	1.4145	0.0970	1.8972	9.2356
ES	137	2.7935	2.0224	-1.0686	2.7533	10.0594

FI	137	1.8474	1.7420	-1.0385	1.3885	8.8654
FR	137	1.7076	1.1768	-0.4232	1.6943	6.0668
GB	137	2.6481	1.7979	0.3339	2.2888	9.4097
GR	138	4.2411	4.9626	-2.3797	3.1253	21.7326
HU	138	9.4805	8.9596	-1.0588	5.5383	37.1026
IT	137	2.5280	2.0507	-0.4847	2.1537	11.7462
NL	138	2.3862	1.7125	0.0033	2.1289	12.2630
NO	138	2.4019	1.2807	-1.4268	2.2680	6.7409
PT	138	2.9791	2.7849	-1.5130	2.5249	13.3373
SE	137	2.0817	2.5188	-1.4166	1.5857	11.5566

Note. Values are in percent.

Table A7

Descriptive Statistics – Population (level)

Country	N	Mean	SD	Min	Median	Max
AT	141	8.370e+06	423168.3766	7.678e+06	8.308e+06	9.178e+06
BE	141	1.079e+07	602574.2852	9.967e+06	1.067e+07	1.186e+07
CH	141	7.756e+06	686261.2142	6.716e+06	7.599e+06	9.006e+06
DE	141	8.202e+07	961118.9029	7.943e+07	8.216e+07	8.352e+07
DK	141	5.519e+06	249133.6175	5.141e+06	5.478e+06	5.977e+06
ES	141	4.403e+07	3.351e+06	3.887e+07	4.559e+07	4.885e+07
FI	141	5.316e+06	184363.5779	4.986e+06	5.301e+06	5.620e+06
FR	141	6.376e+07	3.318e+06	5.826e+07	6.420e+07	6.855e+07
GB	141	6.219e+07	3.759e+06	5.725e+07	6.156e+07	6.923e+07
GR	141	1.075e+07	246128.8098	1.020e+07	1.076e+07	1.112e+07
HU	141	1.001e+07	264769.3374	9.562e+06	1.005e+07	1.037e+07
IT	141	5.846e+07	1.354e+06	5.672e+07	5.895e+07	6.032e+07
NL	141	1.646e+07	828737.5248	1.495e+07	1.641e+07	1.799e+07
NO	141	4.835e+06	426108.6698	4.241e+06	4.739e+06	5.572e+06
PT	141	1.034e+07	212018.0961	9.952e+06	1.037e+07	1.069e+07
SE	141	9.408e+06	642315.3272	8.559e+06	9.184e+06	1.057e+07

Note. Population in absolute persons. Quarter-interpolated geometrically from annual World Bank values.

Table A8

Descriptive Statistics – 10-Year Government Bond Yield (%)

Country	N	Mean	SD	Min	Median	Max
AT	145	3.7874	2.4788	-0.4106	3.8164	8.8533
BE	145	4.0148	2.6333	-0.3720	3.9997	10.1513
CH	145	2.2821	2.0348	-0.7763	2.3283	6.7767
DE	145	3.4926	2.5615	-0.6135	3.4771	8.9483
DK	145	3.8733	2.9098	-0.4927	3.6613	11.0153
ES	145	5.0135	3.5377	0.1022	4.1717	14.7533
FI	145	4.2463	3.4449	-0.3967	3.8033	13.3733
FR	145	3.8827	2.5768	-0.3200	3.6433	10.1767
GB	145	4.5514	2.7319	0.2514	4.4623	12.3167
GR	116	6.2491	4.4539	0.6967	4.9650	25.4000
HU	109	6.2538	2.2684	1.9233	6.8533	10.4200
IT	141	5.1444	3.3309	0.6367	4.2970	13.8480
NL	145	3.6596	2.5198	-0.5413	3.7393	9.1387
NO	145	4.5176	2.5185	0.6067	4.0600	10.8333
PT	131	4.7762	2.9482	0.0933	4.2167	13.2233
SE	145	4.2262	3.3928	-0.1664	3.4300	13.4400

Note. Values are in percent.

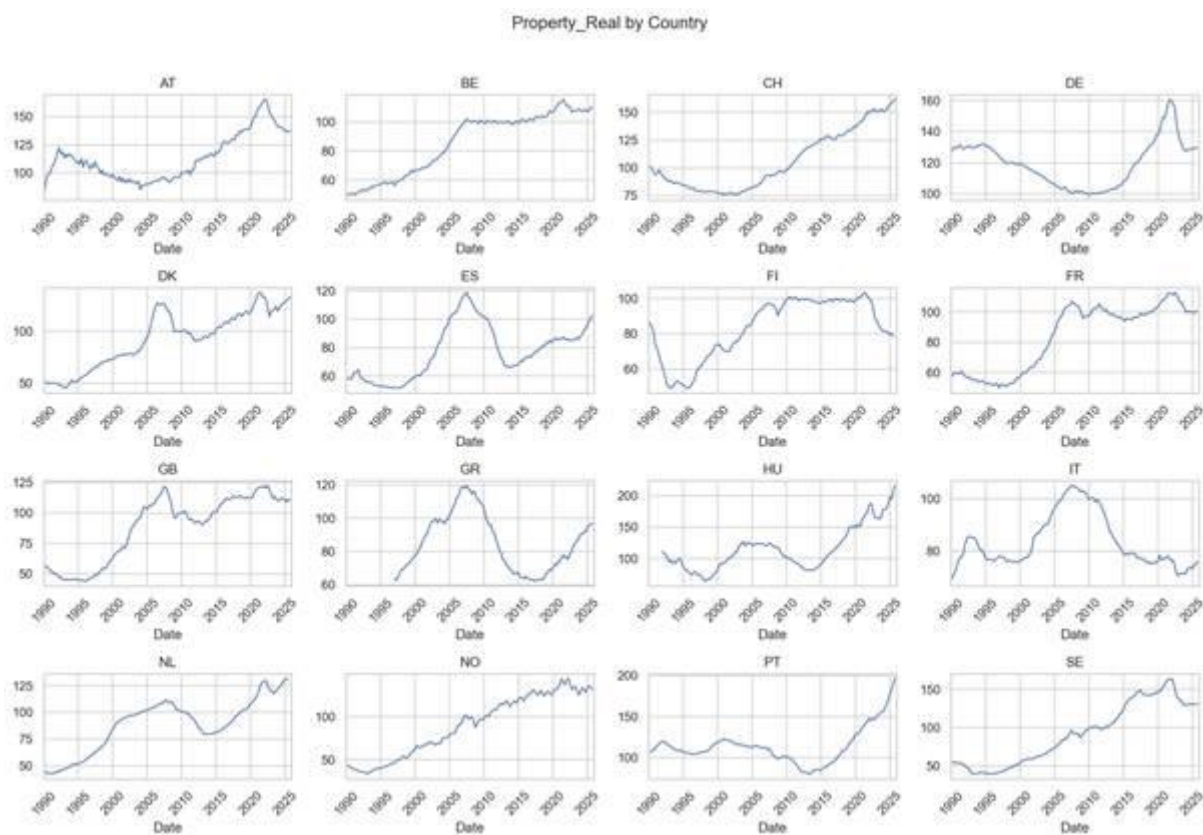
Appendix B

Exploratory Time-Series Plots

This appendix shows the level series for each of the six core variables across all 16 countries in the panel. Each subplot uses the country's native unit and full available sample. These plots motivate the unit-root testing in Appendix C: most series exhibit either a clear stochastic trend (house prices, GDP per capita, population) or a slow mean-reverting pattern with persistent regime shifts (inflation, long-term yields), supporting the $I(1)$ treatment that underlies the cointegration analysis.

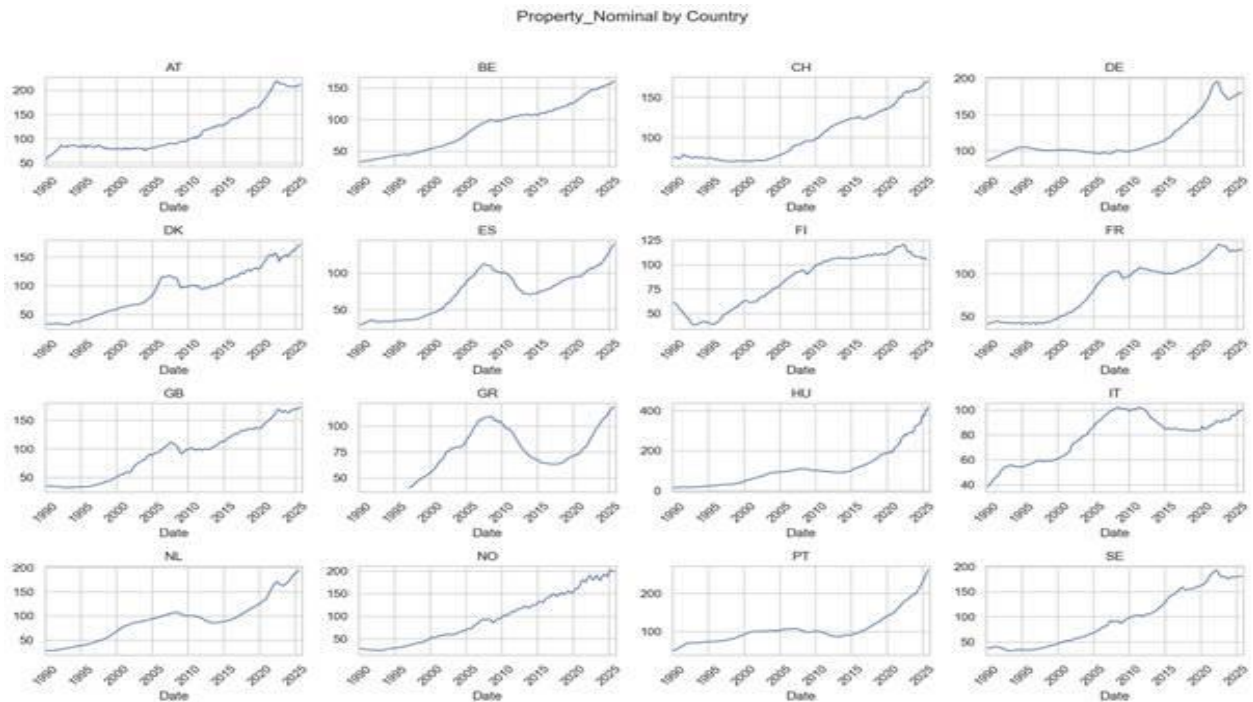
Figure B1

Real Residential Property Price Index by Country



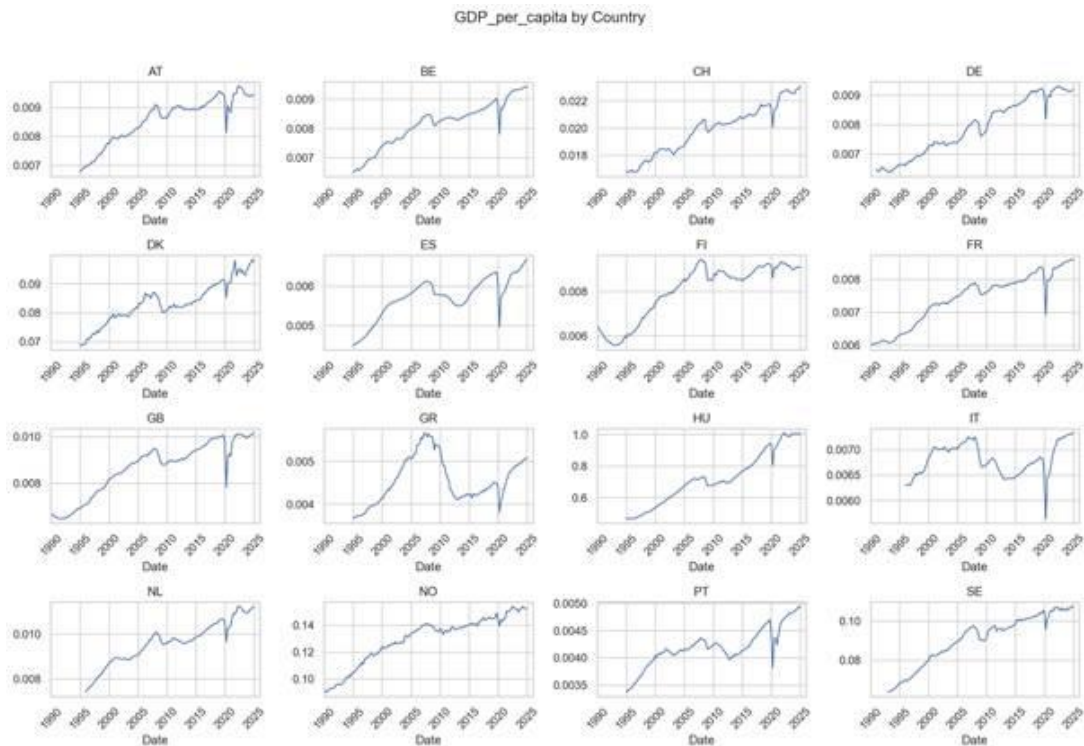
Note. Quarterly observations 1990Q1–2025Q4 where available. Vertical scales are country-specific; the panel layout intentionally preserves within-country amplitude. Source: BIS Residential Property Price Statistics via FRED.

Figure B2
Nominal Residential Property Price Index by Country



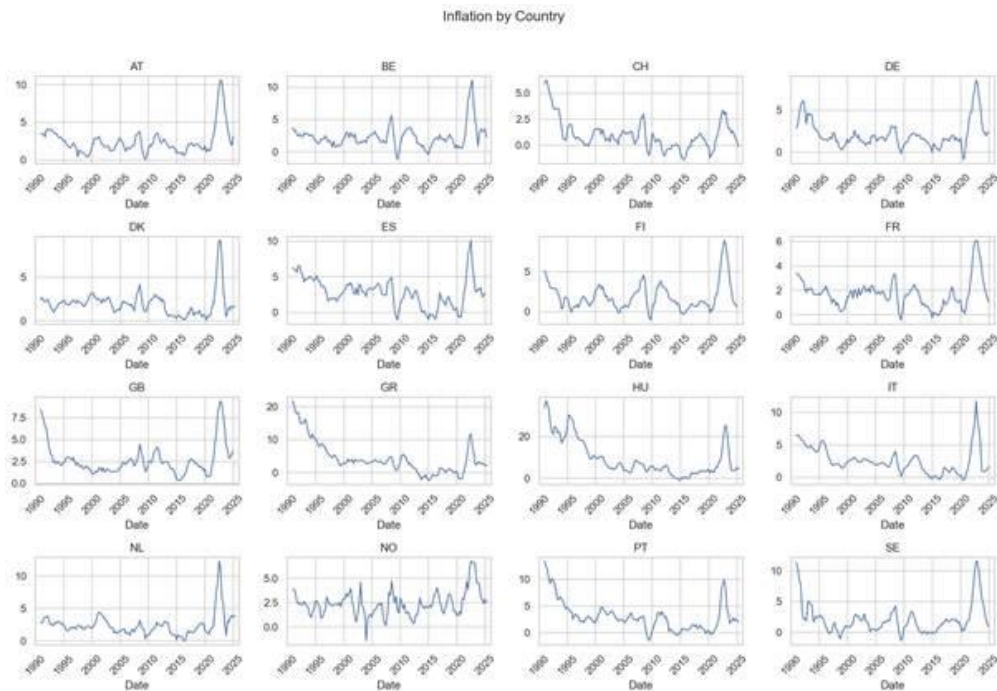
Note. Quarterly observations 1990Q1–2025Q4 where available. Vertical scales are country-specific; the panel layout intentionally preserves within-country amplitude. Source: BIS Residential Property Price Statistics via FRED.

Figure B3
Real GDP per Capita by Country



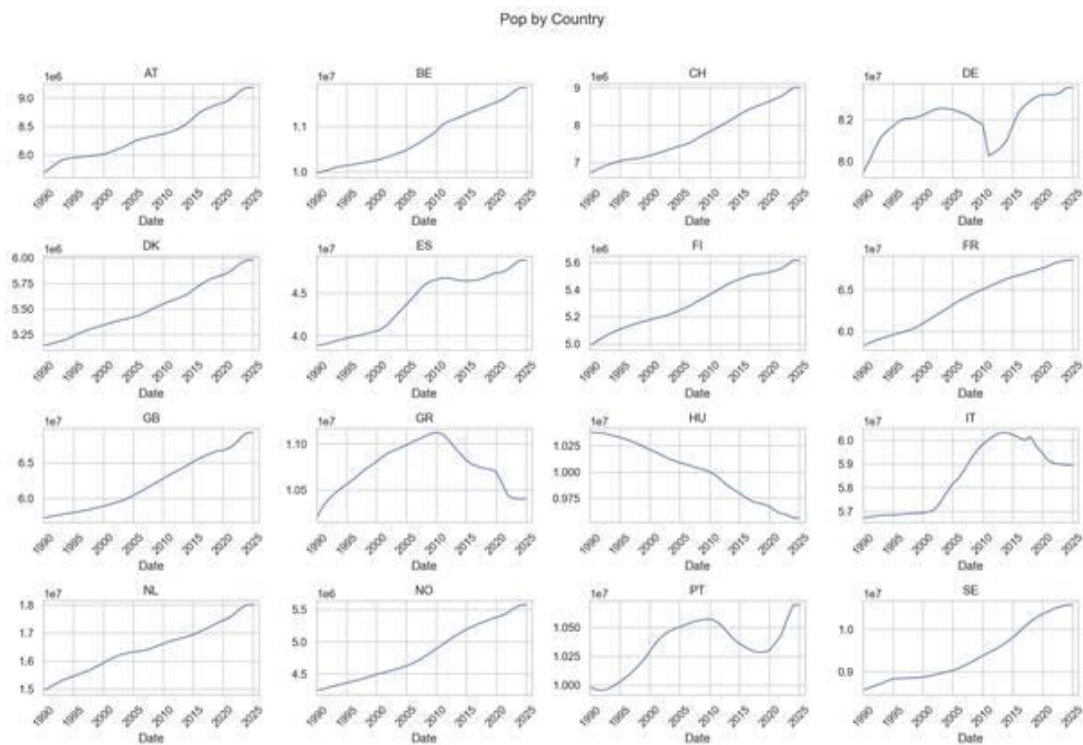
Note. Quarterly observations 1990Q1–2025Q4 where available. Vertical scales are country-specific; the panel layout intentionally preserves within-country amplitude. Source: FRED national accounts series; population from World Bank.

Figure B4
CPI Inflation (Year-over-Year, %) by Country



Note. Quarterly observations 1990Q1–2025Q4 where available. Vertical scales are country-specific; the panel layout intentionally preserves within-country amplitude. Source: FRED CPI series, year-over-year change.

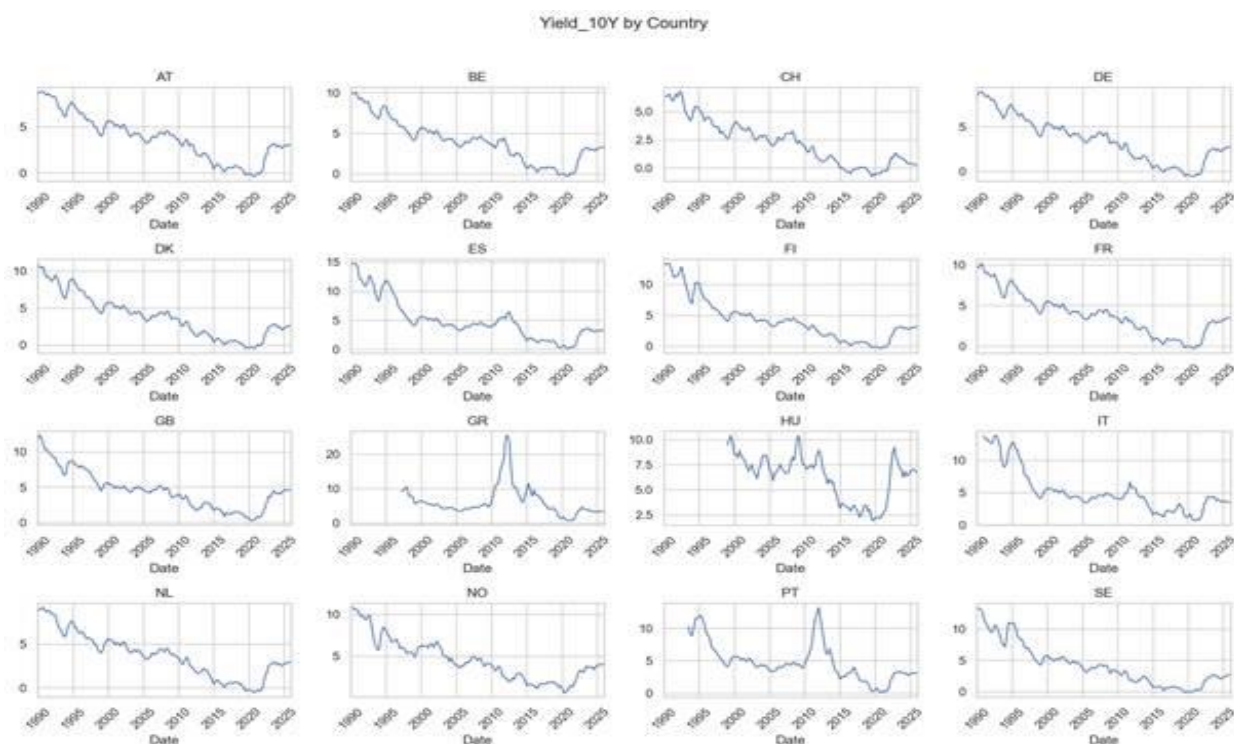
Figure B5
Total Population by Country



Note. Quarterly observations 1990Q1–2025Q4 where available. Vertical scales are country-specific; the panel layout intentionally preserves within-country amplitude. Source: World Bank Population, total.

Figure B6

10-Year Government Bond Yield (%) by Country



Note. Quarterly observations 1990Q1–2025Q4 where available. Vertical scales are country-specific; the panel layout intentionally preserves within-country amplitude. Source: FRED IRLTLT01 series, quarterly mean.

Appendix C

Augmented Dickey–Fuller Test Results

This appendix reports the full ADF unit-root tests underlying the $I(1)$ filter described in §4.6. For each country–variable pair, two ADF tests are run: one on the level (with constant and trend for trending variables, constant only for inflation and yield) and one on the first difference (constant only). The lag length is selected by AIC up to a maximum of 12 lags. A variable enters the country-level VECM as an $I(1)$ candidate only if the level test fails to reject the unit-root null at 5% ($p > 0.05$) AND the first-difference test rejects it ($p \leq 0.05$). The house-price series is always retained as the target variable, irrespective of its ADF outcome – cases in which it fails the $I(1)$ filter are flagged in §6.8 as a limitation.

Table C1

Augmented Dickey–Fuller Tests in Levels and First Differences

Country	Variable	ADF (lev)	p (lev)	ADF (Δ)	p (Δ)	$I(1)$?
AT	L_Property_Real	-1.350	0.8750	-4.528	0.0002	Yes
AT	L_Property_Nominal	-1.042	0.9381	-4.414	0.0003	Yes
AT	L_GDP_per_capita	-2.351	0.4058	-5.376	3.803e-06	Yes
AT	L_Pop	-2.397	0.3811	-2.491	0.1177	No

AT	Inflation	-0.986	0.7584	-7.242	1.877e-10	Yes
AT	Yield_10Y	-2.134	0.2311	-8.332	0.0000	Yes
BE	L_Property_Real	-1.320	0.8828	-3.129	0.0245	Yes
BE	L_Property_Nominal	-2.159	0.5127	-2.614	0.0901	No
BE	L_GDP_per_capita	-2.680	0.2446	-14.282	0.0000	Yes
BE	L_Pop	-2.444	0.3567	-2.029	0.2739	No
BE	Inflation	-2.432	0.1330	-6.968	8.808e-10	Yes
BE	Yield_10Y	-2.244	0.1907	-8.168	0.0000	Yes
CH	L_Property_Real	-2.559	0.2992	-3.843	0.0025	Yes
CH	L_Property_Nominal	-1.949	0.6288	-3.866	0.0023	Yes
CH	L_GDP_per_capita	-3.095	0.1074	-11.364	0.0000	Yes
CH	L_Pop	-2.955	0.1451	-1.875	0.3439	No
CH	Inflation	-3.783	0.0031	-5.115	1.309e-05	No
CH	Yield_10Y	-1.867	0.3476	-8.617	0.0000	Yes
DE	L_Property_Real	-1.481	0.8357	-4.336	0.0004	Yes
DE	L_Property_Nominal	-1.604	0.7908	-2.668	0.0798	No
DE	L_GDP_per_capita	-3.281	0.0695	-13.083	0.0000	Yes
DE	L_Pop	-2.600	0.2799	-2.674	0.0787	No
DE	Inflation	-2.037	0.2705	-7.220	2.129e-10	Yes
DE	Yield_10Y	-2.223	0.1982	-8.369	0.0000	Yes
DK	L_Property_Real	-2.220	0.4786	-2.708	0.0726	No
DK	L_Property_Nominal	-2.165	0.5098	-2.645	0.0841	No
DK	L_GDP_per_capita	-2.289	0.4403	-13.377	0.0000	Yes
DK	L_Pop	-1.510	0.8258	-2.185	0.2116	No
DK	Inflation	-3.459	0.0091	-4.587	0.0001	No
DK	Yield_10Y	-2.223	0.1979	-7.467	5.172e-11	Yes
ES	L_Property_Real	-3.465	0.0432	-2.460	0.1256	No
ES	L_Property_Nominal	-3.269	0.0715	-1.655	0.4543	No
ES	L_GDP_per_capita	-2.764	0.2104	-13.107	0.0000	Yes
ES	L_Pop	-1.340	0.8777	-2.060	0.2610	No
ES	Inflation	-2.262	0.1846	-4.951	2.776e-05	Yes
ES	Yield_10Y	-2.870	0.0489	-7.168	2.851e-10	No
FI	L_Property_Real	-1.653	0.7708	-2.972	0.0376	Yes

FI	L_Property_Nominal	-0.437	0.9857	-2.997	0.0352	Yes
FI	L_GDP_per_capita	-1.484	0.8345	-4.391	0.0003	Yes
FI	L_Pop	-2.654	0.2555	-2.556	0.1023	No
FI	Inflation	-3.187	0.0208	-5.726	6.754e-07	No
FI	Yield_10Y	-3.058	0.0299	-3.996	0.0014	No
FR	L_Property_Real	-2.232	0.4720	-1.707	0.4277	No
FR	L_Property_Nominal	-2.364	0.3991	-2.322	0.1648	No
FR	L_GDP_per_capita	-2.162	0.5113	-10.795	0.0000	Yes
FR	L_Pop	-0.417	0.9864	-0.515	0.8891	No
FR	Inflation	-1.618	0.4736	-5.610	1.210e-06	Yes
FR	Yield_10Y	-2.266	0.1833	-8.058	1.670e-12	Yes
GB	L_Property_Real	-2.433	0.3621	-2.163	0.2199	No
GB	L_Property_Nominal	-2.764	0.2106	-2.169	0.2178	No
GB	L_GDP_per_capita	-1.723	0.7405	-10.737	0.0000	Yes
GB	L_Pop	-2.820	0.1899	-1.595	0.4860	No
GB	Inflation	-1.969	0.3002	-6.274	3.931e-08	Yes
GB	Yield_10Y	-3.100	0.0266	-7.624	2.090e-11	No
GR	L_Property_Real	-2.742	0.2191	-2.005	0.2846	No
GR	L_Property_Nominal	-3.560	0.0334	-1.873	0.3449	No
GR	L_GDP_per_capita	-1.749	0.7287	-6.022	1.489e-07	Yes
GR	L_Pop	-1.327	0.8811	-1.788	0.3864	No
GR	Inflation	-3.052	0.0303	-3.788	0.0030	No
GR	Yield_10Y	-2.481	0.1202	-6.366	2.406e-08	Yes
HU	L_Property_Real	-2.192	0.4941	-2.195	0.2082	No
HU	L_Property_Nominal	-3.109	0.1042	-1.400	0.5822	No
HU	L_GDP_per_capita	-1.799	0.7052	-13.151	0.0000	Yes
HU	L_Pop	-2.403	0.3781	-2.464	0.1245	No
HU	Inflation	-2.275	0.1801	-3.106	0.0261	Yes
HU	Yield_10Y	-2.425	0.1349	-7.088	4.496e-10	Yes
IT	L_Property_Real	-1.555	0.8094	-3.449	0.0094	Yes
IT	L_Property_Nominal	-1.326	0.8813	-3.551	0.0068	Yes
IT	L_GDP_per_capita	-2.372	0.3944	-11.905	0.0000	Yes
IT	L_Pop	-1.750	0.7285	-1.944	0.3117	No

IT	Inflation	-1.978	0.2964	-5.158	1.068e-05	Yes
IT	Yield_10Y	-2.580	0.0971	-6.919	1.163e-09	Yes
NL	L_Property_Real	-2.170	0.5068	-2.350	0.1564	No
NL	L_Property_Nominal	-2.327	0.4192	-1.863	0.3498	No
NL	L_GDP_per_capita	-3.475	0.0422	-11.137	0.0000	No
NL	L_Pop	-2.378	0.3915	-2.531	0.1082	No
NL	Inflation	-1.855	0.3536	-6.710	3.702e-09	Yes
NL	Yield_10Y	-2.211	0.2022	-7.885	4.588e-12	Yes
NO	L_Property_Real	-0.766	0.9684	-4.113	0.0009	Yes
NO	L_Property_Nominal	-1.109	0.9276	-4.355	0.0004	Yes
NO	L_GDP_per_capita	-1.860	0.6751	-14.290	0.0000	Yes
NO	L_Pop	-1.878	0.6662	-1.032	0.7413	No
NO	Inflation	-2.757	0.0647	-5.234	7.478e-06	Yes
NO	Yield_10Y	-2.554	0.1030	-8.164	0.0000	Yes
PT	L_Property_Real	0.111	0.9952	-2.924	0.0426	Yes
PT	L_Property_Nominal	-0.495	0.9836	-2.525	0.1096	No
PT	L_GDP_per_capita	-2.621	0.2705	-6.323	3.042e-08	Yes
PT	L_Pop	-2.156	0.5144	-2.697	0.0745	No
PT	Inflation	-4.320	0.0004	-4.559	0.0002	No
PT	Yield_10Y	-2.441	0.1306	-6.018	1.521e-07	Yes
SE	L_Property_Real	-2.153	0.5164	-4.174	0.0007	Yes
SE	L_Property_Nominal	-2.655	0.2554	-4.049	0.0012	Yes
SE	L_GDP_per_capita	-2.001	0.6009	-12.409	0.0000	Yes
SE	L_Pop	-2.556	0.3004	-1.481	0.5430	No
SE	Inflation	-2.864	0.0497	-5.247	7.030e-06	No
SE	Yield_10Y	-2.701	0.0739	-6.587	7.279e-09	Yes

Note. ADF (lev) is the test statistic on the level series; ADF (Δ) on the first difference. Critical values are not tabulated here; the p-values follow MacKinnon (1996) and are produced by statsmodels.tsa.stattools.adfuller. "I(1)?" = Yes when $p(\text{lev}) > 0.05$ and $p(\Delta) \leq 0.05$. L_ prefix denotes natural-log levels.

Appendix D

Cointegration and Lag Selection

This appendix reports the Johansen trace test for cointegration and the AIC-based lag selection underlying the VECM specifications. Both the real and the nominal house-price specification are estimated for each country to enable the robustness comparison in Appendix G. The set of endogenous variables per country is determined by the $I(1)$ filter (Appendix C); only variables that pass the filter enter the test. The VECM is estimated only when the trace test identifies at least one cointegrating rank at the 5% level.

Table D1
Johansen Trace Test Results and VECM Eligibility

Country	Spec	Endogenous variables	Sample	N	k_Δ	Rank	VECM
AT	Nominal	HP_N;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	1	2	Yes
AT	Real	HP_R;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	1	2	Yes
BE	Nominal	HP_N;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	3	3	Yes
BE	Real	HP_R;GDP_pc;Inflation;Yield	1995Q1–2025Q1	121	3	1	Yes
CH	Nominal	HP_N;GDP_pc;Yield	1995Q1–2025Q1	121	1	1	Yes
CH	Real	HP_R;GDP_pc;Yield	1995Q1–2025Q1	121	1	1	Yes
DE	Nominal	HP_N;GDP_pc;Inflation;Yield	1991Q1–2025Q1	137	5	4	Yes
DE	Real	HP_R;GDP_pc;Inflation;Yield	1991Q1–2025Q1	137	5	3	Yes
DK	Nominal	HP_N;GDP_pc;Yield	1995Q1–2025Q1	121	3	0	No
DK	Real	HP_R;GDP_pc;Yield	1995Q1–2025Q1	121	3	0	No
ES	Nominal	HP_N;GDP_pc;Inflation	1995Q1–2025Q1	121	4	3	Yes
ES	Real	HP_R;GDP_pc;Inflation	1995Q1–2025Q1	121	4	3	Yes
FI	Nominal	HP_N;GDP_pc	1990Q1–2025Q1	141	2	2	Yes
FI	Real	HP_R;GDP_pc	1990Q1–2025Q1	141	2	2	Yes

FR	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	5	1	Yes
FR	Real	HP_R;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	5	2	Yes
GB	Nomi- nal	HP_N;GDP_pc;Inflation	1991Q1– 2025Q1	137	4	3	Yes
GB	Real	HP_R;GDP_pc;Inflation	1991Q1– 2025Q1	137	4	3	Yes
GR	Nomi- nal	HP_N;GDP_pc;Yield	1997Q2– 2025Q1	112	1	1	Yes
GR	Real	HP_R;GDP_pc;Yield	1997Q2– 2025Q1	112	1	1	Yes
HU	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1999Q1– 2025Q1	105	1	1	Yes
HU	Real	HP_R;GDP_pc;Inflation;Yield	1999Q1– 2025Q1	105	4	0	No
IT	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1996Q1– 2025Q1	117	3	4	Yes
IT	Real	HP_R;GDP_pc;Inflation;Yield	1996Q1– 2025Q1	117	3	2	Yes
NL	Nomi- nal	HP_N;Inflation;Yield	1991Q1– 2025Q2	138	3	3	Yes
NL	Real	HP_R;Inflation;Yield	1991Q1– 2025Q2	138	3	3	Yes
NO	Nomi- nal	HP_N;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	4	4	Yes
NO	Real	HP_R;GDP_pc;Inflation;Yield	1991Q1– 2025Q1	137	4	4	Yes
PT	Nomi- nal	HP_N;GDP_pc;Yield	1995Q1– 2025Q1	121	1	2	Yes
PT	Real	HP_R;GDP_pc;Yield	1995Q1– 2025Q1	121	1	1	Yes
SE	Nomi- nal	HP_N;GDP_pc;Yield	1993Q1– 2025Q1	129	1	3	Yes
SE	Real	HP_R;GDP_pc;Yield	1993Q1– 2025Q1	129	4	0	No

Note. Spec = real or nominal house-price specification. Endogenous variables: HP_R = log real house price, HP_N = log nominal house price, GDP_pc = log GDP per capita, Inflation = CPI YoY%, Yield = 10Y bond yield. k_{Δ} is the lag order in differences chosen by AIC (max 6). Rank is the cointegrating rank from the Johansen trace test at the 5% level. "VECM = Yes" requires Rank ≥ 1 and $N \geq 40$.

Table D2

Lag Length Selection (AIC, Max 6 Quarters)

Country	Spec	VAR lag	k_ar_diff (VECM)	N obs
AT	Nominal	1	1	121
AT	Real	1	1	121
BE	Nominal	4	3	121
BE	Real	4	3	121
CH	Nominal	1	1	121
CH	Real	1	1	121
DE	Nominal	6	5	137
DE	Real	6	5	137
DK	Nominal	4	3	121
DK	Real	4	3	121
ES	Nominal	5	4	121
ES	Real	5	4	121
FI	Nominal	3	2	141
FI	Real	3	2	141
FR	Nominal	6	5	137
FR	Real	6	5	137
GB	Nominal	5	4	137
GB	Real	5	4	137
GR	Nominal	2	1	112
GR	Real	2	1	112
HU	Nominal	2	1	105
HU	Real	5	4	105
IT	Nominal	4	3	117
IT	Real	4	3	117
NL	Nominal	4	3	138
NL	Real	4	3	138
NO	Nominal	5	4	137
NO	Real	5	4	137
PT	Nominal	2	1	121
PT	Real	2	1	121
SE	Nominal	2	1	129

SE	Real	5	4	129
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Note. VAR lag is the AIC-optimal lag order over levels; k_ar_diff is the corresponding lag count in the VECM differenced form, computed as $\max(\text{VAR lag} - 1, 1)$ to enforce a technical minimum of k_ar_diff = 1. For countries where the AIC-optimal VAR lag is 1 (AT, CH), the floor binds and k_ar_diff is set to 1.

Appendix E

VECM Coefficient Estimates and Diagnostics

This appendix reports the full set of estimated cointegration loadings (β) and adjustment coefficients (α) for the first cointegrating vector (CE_0) in each VECM specification, together with the diagnostic statistics that drive the Solid/Weak classification used throughout §5–§6. β -coefficients are normalised by statsmodels' identity-block convention; entries that are pinned to 0.0 by this normalisation are reported as such. Variables that do not enter the country-specific system (after $I(1)$ filtering) are shown as blanks. The diagnostic table reports the Bonferroni-corrected LM threshold $0.05/k$, the worst-equation LM p-value, and the maximum modulus of the AR-companion roots.

Table E1

Cointegration Loadings (β) on the First Cointegration Vector

Country	Spec	k_Δ	Rank	β_{HP}	β_{GDP}	β_{Infl}	β_{Yield}
AT	Nominal	1	2	1.0000	0.0000	-0.1799	0.2203
AT	Real	1	2	1.0000	0.0000	0.0818	0.2204
BE	Nominal	3	3	1.0000	0.0000	0.0000	0.0611
BE	Real	3	1	1.0000	-4.2644	0.3036	-0.0800
CH	Nominal	1	1	1.0000	-3.4308		0.0390
CH	Real	1	1	1.0000	-2.2545		0.0817
DE	Nominal	5	4	1.0000	0.0000	0.0000	0.0000
DE	Real	5	3	1.0000	0.0000	0.0000	-0.1395
ES	Nominal	4	3	1.0000	0.0000	0.0000	
ES	Real	4	3	1.0000	0.0000	0.0000	
FI	Nominal	2	2	1.0000	0.0000		
FI	Real	2	2	1.0000	0.0000		
FR	Nominal	5	1	1.0000	-4.2462	-0.1657	0.0221
FR	Real	5	2	1.0000	0.0000	0.2125	0.0503
GB	Nominal	4	3	1.0000	0.0000	0.0000	
GB	Real	4	3	1.0000	0.0000	0.0000	
GR	Nominal	1	1	1.0000	-0.5706		0.1122
GR	Real	1	1	1.0000	-3.0101		-0.0844
HU	Nominal	1	1	1.0000	-22.2051	1.5684	-2.7748
IT	Nominal	3	4	1.0000	0.0000	0.0000	0.0000

IT	Real	3	2	1.0000	0.0000	0.3559	-0.1419
NL	Nominal	3	3	1.0000		0.0000	0.0000
NL	Real	3	3	1.0000		0.0000	0.0000
NO	Nominal	4	4	1.0000	0.0000	0.0000	0.0000
NO	Real	4	4	1.0000	0.0000	0.0000	0.0000
PT	Nominal	1	2	1.0000	0.0000		-0.5730
PT	Real	1	1	1.0000	-3.3932		-0.4751
SE	Nominal	1	3	1.0000	0.0000		0.0000

Note. β_{HP} is the loading on the house-price variable in CE_0 (typically normalised to 1 by the statsmodels identity-block convention). Blank cells indicate variables that were removed from the country-specific system by the I(1) filter (Appendix C) or normalised to 0 by construction. Values reported in scientific notation reflect numerical noise from the identity-block normalisation rather than estimated coefficients.

Table E2

Table E2: VECM Adjustment Coefficients (α) on the First Cointegration Vector

Country	Spec	α_{HP}	α_{GDP}	α_{Infl}	α_{Yield}
AT	Nominal	-0.0091	0.0120	0.7182	-0.2155
AT	Real	-0.0292	0.0144	1.1025	-0.4480
BE	Nominal	-0.0685	0.0593	1.6827	0.1923
BE	Real	-0.0052	-0.0051	-1.1711	0.1614
CH	Nominal	-0.0437	0.0216		-0.1510
CH	Real	-0.0534	0.0151		-0.1672
DE	Nominal	-0.0015	-0.0033	1.1449	0.4145
DE	Real	0.0028	-0.0159	0.6627	0.0356
ES	Nominal	-0.0200	0.0287	1.0705	
ES	Real	-0.0348	-0.0062	0.6729	
FI	Nominal	-0.0443	0.0075		
FI	Real	-0.0585	-0.0086		
FR	Nominal	-0.0042	0.0227	0.7253	-0.1751
FR	Real	-0.0369	0.0065	0.6676	-0.0735
GB	Nominal	-0.0093	0.0683	1.0647	
GB	Real	-0.0445	0.0399	0.7233	
GR	Nominal	-0.0125	-0.0096		-0.6643
GR	Real	0.0388	0.0254		1.1984

HU	Nominal	0.0012	0.0004	-0.1298	-0.0014
IT	Nominal	-0.0286	0.0064	0.5351	-0.0015
IT	Real	-0.0013	0.0064	-0.8034	0.0534
NL	Nominal	-0.0062		0.6413	0.0050
NL	Real	-0.0135		0.6716	0.0679
NO	Nominal	-0.0396	0.0195	0.1765	-0.0419
NO	Real	-0.0886	0.0137	-0.1089	-0.0122
PT	Nominal	0.0105	0.0525		0.1277
PT	Real	0.0050	0.0034		0.1811
SE	Nominal	-0.0300	0.0104		0.0446

Note. α_{HP} is the speed-of-adjustment loading on the house-price equation. Negative values indicate that house prices themselves move toward the long-run equilibrium following a deviation; positive or near-zero values indicate that adjustment occurs primarily through other system variables. Blank cells indicate variables removed by the $I(1)$ filter.

Table E3

VECM Diagnostics and Solid/Weak Classification

Country	Spec	k_eq	Bonf. thr.	LM p (min)	No autocorr.	Max root	Stable	Evaluation
AT	Nominal	4	0.0125	0.1867	Yes	1.0000	Yes	Solid
AT	Real	4	0.0125	0.2253	Yes	1.0000	Yes	Solid
BE	Nominal	4	0.0125	1.153e-06	No	1.0000	Yes	Weak
BE	Real	4	0.0125	1.030e-06	No	1.0000	Yes	Weak
CH	Nominal	3	0.0167	0.1382	Yes	1.0000	Yes	Solid
CH	Real	3	0.0167	0.2137	Yes	1.0000	Yes	Solid
DE	Nominal	4	0.0125	0.2567	Yes	0.9909	Yes	Solid
DE	Real	4	0.0125	0.2791	Yes	1.0000	Yes	Solid
ES	Nominal	3	0.0167	0.0486	Yes	0.9769	Yes	Solid
ES	Real	3	0.0167	0.2835	Yes	0.9829	Yes	Solid
FI	Nominal	2	0.0250	0.0570	Yes	0.9777	Yes	Solid
FI	Real	2	0.0250	0.1481	Yes	0.9525	Yes	Solid
FR	Nominal	4	0.0125	0.0016	No	1.0000	Yes	Weak

FR	Real	4	0.0125	0.0113	No	1.0000	Yes	Weak
GB	Nomi- nal	3	0.0167	0.0965	Yes	0.9607	Yes	Solid
GB	Real	3	0.0167	0.0241	Yes	0.9580	Yes	Solid
GR	Nomi- nal	3	0.0167	0.0227	Yes	1.0000	Yes	Solid
GR	Real	3	0.0167	9.717e-08	No	1.0000	Yes	Weak
HU	Nomi- nal	4	0.0125	0.1953	Yes	1.0000	Yes	Solid
IT	Nomi- nal	4	0.0125	0.0025	No	0.9491	Yes	Weak
IT	Real	4	0.0125	0.0073	No	1.0000	Yes	Weak
NL	Nomi- nal	3	0.0167	4.358e-05	No	0.9813	Yes	Weak
NL	Real	3	0.0167	2.258e-05	No	0.9745	Yes	Weak
NO	Nomi- nal	4	0.0125	0.0119	No	0.9830	Yes	Weak
NO	Real	4	0.0125	0.0156	Yes	0.9760	Yes	Solid
PT	Nomi- nal	3	0.0167	0.4066	Yes	1.0000	Yes	Solid
PT	Real	3	0.0167	1.760e-05	No	1.0000	Yes	Weak
SE	Nomi- nal	3	0.0167	0.0362	Yes	0.9851	Yes	Solid

Note. k_{eq} is the number of equations (= dimension of the VECM system). Bonferroni threshold = $0.05 / k_{eq}$, applied to the minimum equation-level LM p-value. "No autocorr." = Yes when min LM p > Bonferroni threshold. Max root is the largest absolute eigenvalue of the AR companion matrix; "Stable" = Yes when Max root $\leq 1 + 10^{-6}$ (numerical tolerance). Solid = stable AND no autocorrelation; Weak = stable AND Max root ≤ 1.01 but autocorrelation present; Not Suitable = Max root > 1.01.

Appendix F

Robustness with Crisis Dummies

This appendix reports the re-estimation of the VECM systems flagged as Solid in Appendix E with two exogenous crisis dummies: the Global Financial Crisis (2008Q1–2009Q4) and the COVID-19 shock (2020Q1–2021Q4). Cointegration rank and lag length are held fixed at the baseline values to isolate the effect of the dummies on the adjustment dynamics. The reported quantity is the speed-of-adjustment coefficient on the house-price equation under crisis-dummy control. Magnitudes within roughly 0.005 of the baseline α (Table E2) indicate that the long-run identification is not driven by the crisis episodes.

Table F1

VECM Re-Estimation with GFC and COVID Crisis Dummies

Country	Spec	k_Δ	Rank	α_{HP} (with dummies)	Status
AT	Nominal	1	2	-0.0141	Estimated
AT	Real	1	2	-0.0384	Estimated
CH	Nominal	1	1	-0.0480	Estimated
CH	Real	1	1	-0.0593	Estimated
DE	Nominal	5	4	-0.0074	Estimated
DE	Real	5	3	0.0032	Estimated
ES	Nominal	4	3	-0.0195	Estimated
ES	Real	4	3	-0.0354	Estimated
FI	Nominal	2	2	-0.0515	Estimated
FI	Real	2	2	-0.0583	Estimated
GB	Nominal	4	3	-0.0303	Estimated
GB	Real	4	3	-0.0400	Estimated
GR	Nominal	1	1	-0.0003	Estimated
HU	Nominal	1	1	0.0061	Estimated
NO	Real	4	4	-0.0908	Estimated
PT	Nominal	1	2	0.0095	Estimated
SE	Nominal	1	3	-0.0324	Estimated

Note. Estimation uses identical lag order and cointegration rank as the baseline VECM (Tables D1, D2). Crisis dummies enter as exogenous regressors only; they do not alter the long-run cointegration structure. The estimated α with crisis dummies is to be compared with the baseline α_{HP} from Table E2 to assess $\Delta\alpha = \alpha_{with_dummies} - \alpha_{baseline}$.

Appendix G

Nominal Specification (Robustness Comparison)

This appendix reports the parallel VECM estimates obtained when the house-price variable is the nominal BIS index rather than the real one. As argued in §4.5, the real specification is preferred on econometric and economic grounds; the nominal results are therefore not used for the headline interpretation but provide a robustness comparison. For Hungary and Sweden the nominal specification is the operative one because the Johansen test fails to identify a real cointegration rank; for the remaining 13 countries, the nominal estimates serve as a cross-check on β -coefficient magnitudes and signs.

Table G1

VECM Estimates – Nominal Specification, All Countries

Country	Spec	Evaluation	LM p (min)	Max root	β_{GDP}	β_{Infl}	β_{Yield}
AT	Nominal	Solid	0.1867	1.0000	0.0000	-0.1799	0.2203
BE	Nominal	Weak	1.153e-06	1.0000	0.0000	0.0000	0.0611
CH	Nominal	Solid	0.1382	1.0000	-3.4308		0.0390
DE	Nominal	Solid	0.2567	0.9909	0.0000	0.0000	0.0000
ES	Nominal	Solid	0.0486	0.9769	0.0000	0.0000	
FI	Nominal	Solid	0.0570	0.9777	0.0000		
FR	Nominal	Weak	0.0016	1.0000	-4.2462	-0.1657	0.0221
GB	Nominal	Solid	0.0965	0.9607	0.0000	0.0000	
GR	Nominal	Solid	0.0227	1.0000	-0.5706		0.1122
HU	Nominal	Solid	0.1953	1.0000	-22.2051	1.5684	-2.7748
IT	Nominal	Weak	0.0025	0.9491	0.0000	0.0000	0.0000
NL	Nominal	Weak	4.358e-05	0.9813		0.0000	0.0000
NO	Nominal	Weak	0.0119	0.9830	0.0000	0.0000	0.0000
PT	Nominal	Solid	0.4066	1.0000	0.0000		-0.5730
SE	Nominal	Solid	0.0362	0.9851	0.0000		0.0000

Note. Specification fixed to nominal house prices for all available countries. Diagnostics follow the same Solid/Weak/Not Suitable convention as Table E3. β -coefficients are reported on the first cointegrating vector with statsmodels normalisation; very small values reflect identity-block normalisation rather than estimated effects.

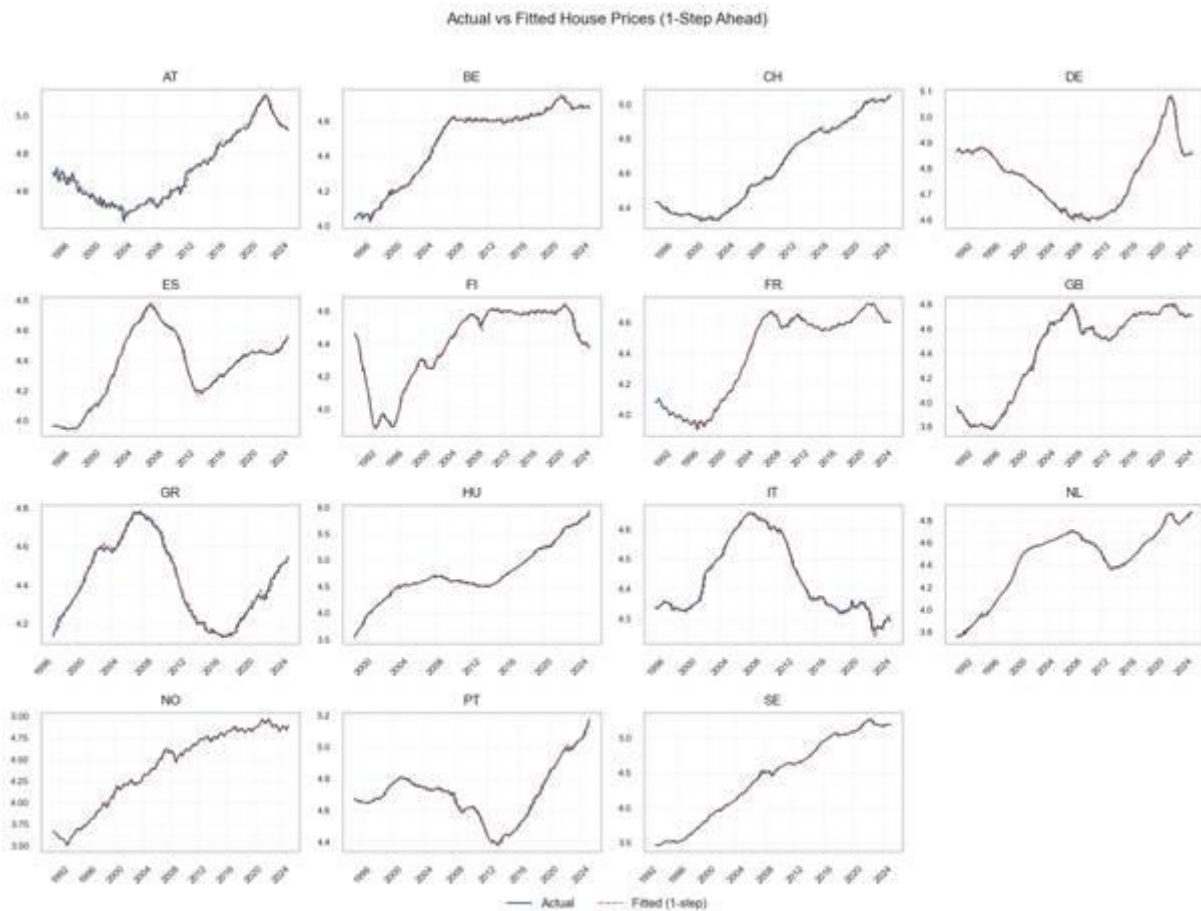
Appendix H

Diagnostic and Forecast Figures

This appendix collects the diagnostic figures discussed in §6.2 and §6.6: the one-step-ahead in-sample fit in levels (Figure H1) and in first differences (Figure H2), the pure- β long-run equilibrium implied by the cointegration vector (Figure H3), the contribution decomposition of annual house-price changes into long-run, persistence, and macro components (Figure H4), and the implied misalignment series $u_t = \beta'X_t - \text{mean}$ (Figure H5). All series are shown for the preferred per-country specification reported in §5.3.

Figure H1

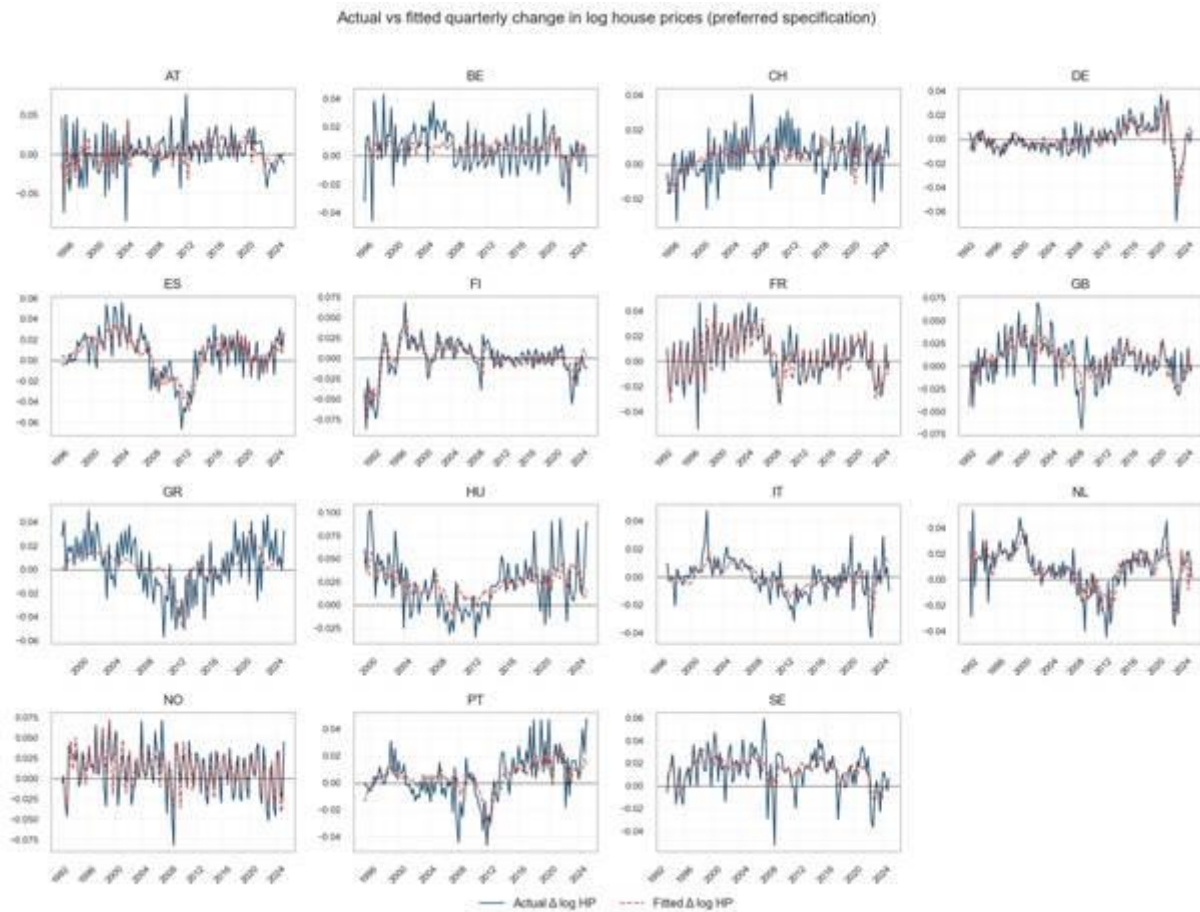
Actual vs. Fitted Log House Prices, One-Step-Ahead (Preferred Specification)



Note. Solid line: realised log house-price level. Dashed line: one-step-ahead VECM fit using the full short-run dynamics (α -correction term plus Γ -lag dynamics). Tight visual fit is constructive – it indicates that conditional one-step prediction performs well – but it is not, on its own, evidence of out-of-sample forecast quality (see §6.2). Country panels use the preferred real or nominal specification per §5.3.

Figure H2

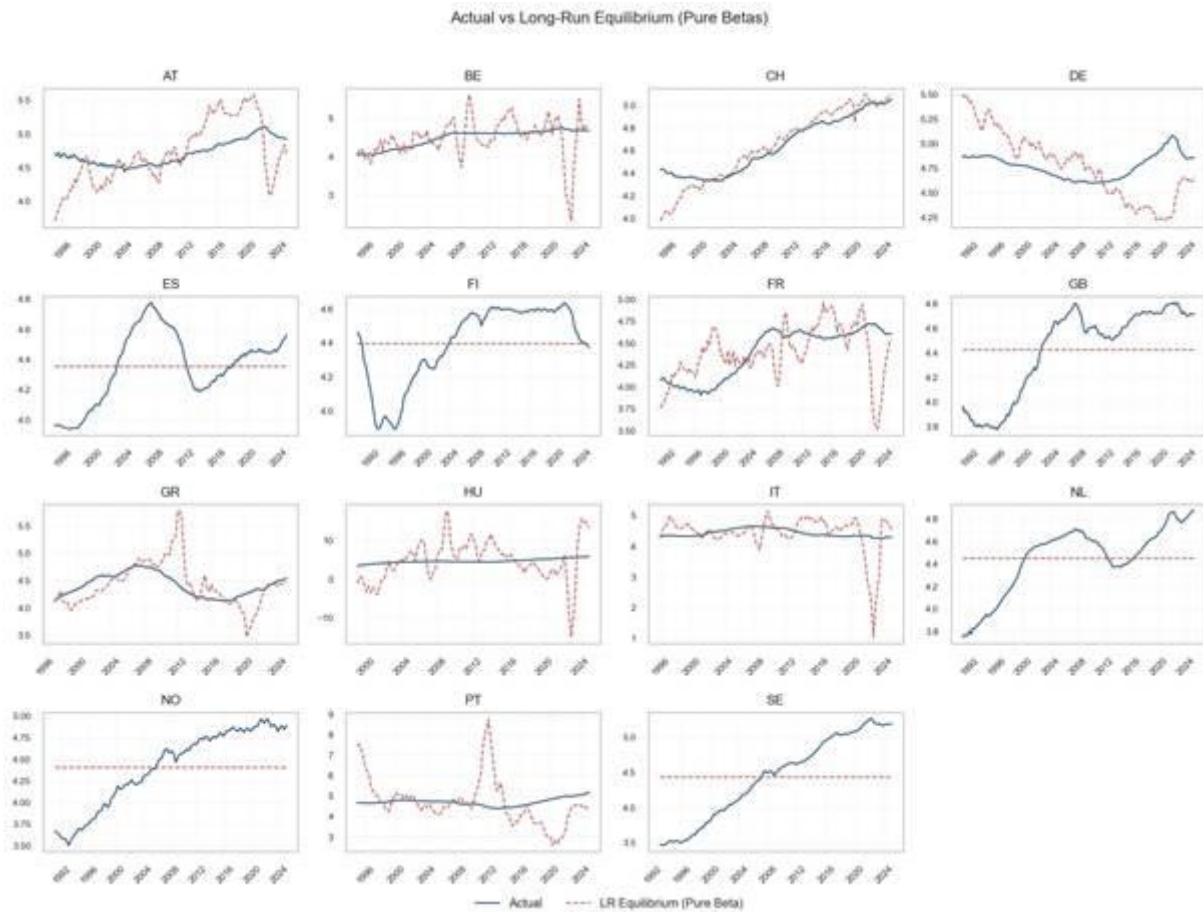
Actual vs. Fitted Quarterly Change in Log House Prices (Preferred Specification)



Note. Solid line: realised quarterly $\Delta \log \text{HP}$. Dashed line: one-step-ahead VECM-implied $\Delta \log \text{HP}$. This differenced view makes the volatility of innovations visible alongside the model's adjustment dynamics. Unlike Figure H3, this plot does not isolate the long-run equilibrium implication of the β -vector.

Figure H3

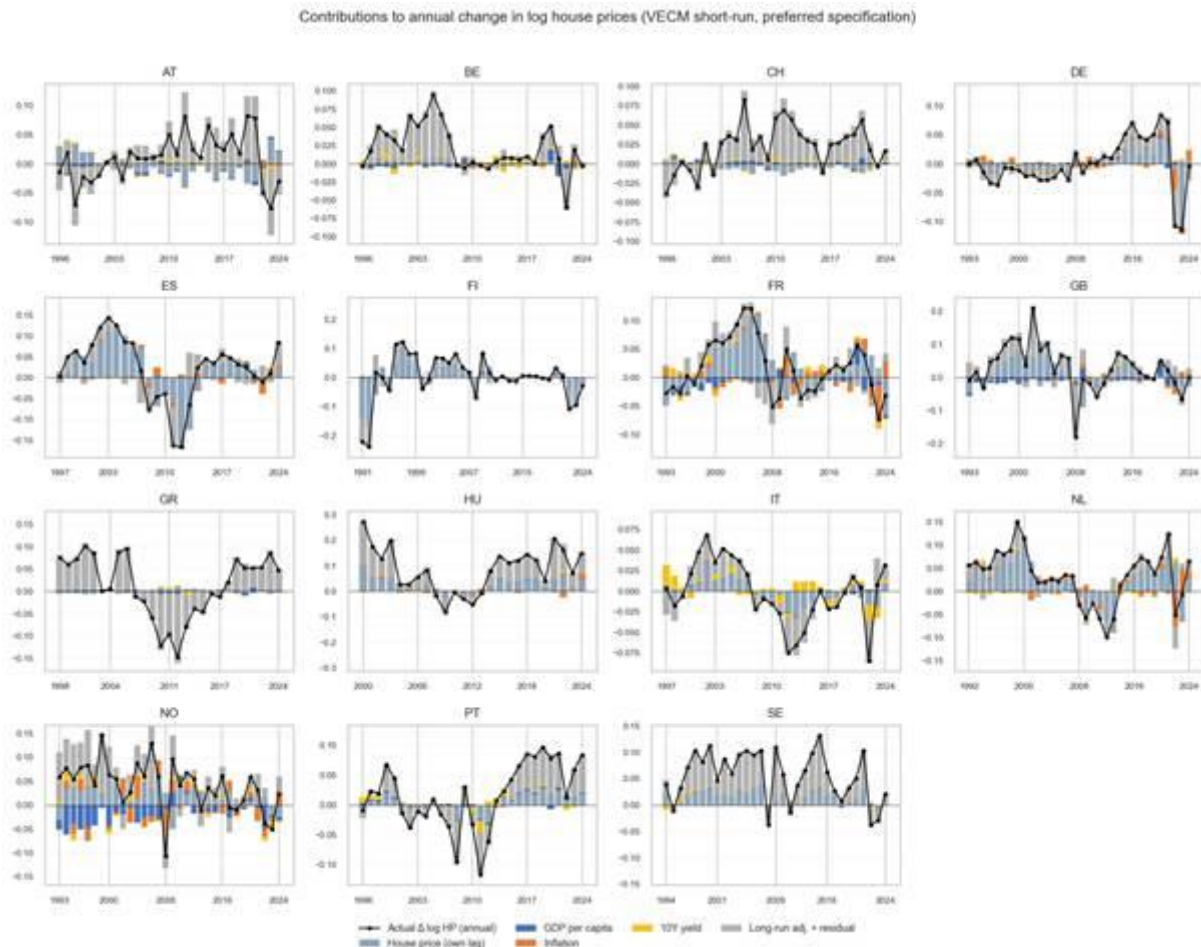
Actual Log House Prices vs. Long-Run Equilibrium Implied by Pure β



Note. Solid line: realised log house-price level. Dashed line: equilibrium level implied by the centred cointegration deviation u_t ($\beta'X_t - \text{country mean}$), with VECM lag dynamics removed. This is the diagnostically decisive plot for evaluating whether the estimated long-run relation tracks the level of the series – the precondition for the economic interpretability of the misalignment series in Figure H5 and the forward-looking projection in §6.5.

Figure H4

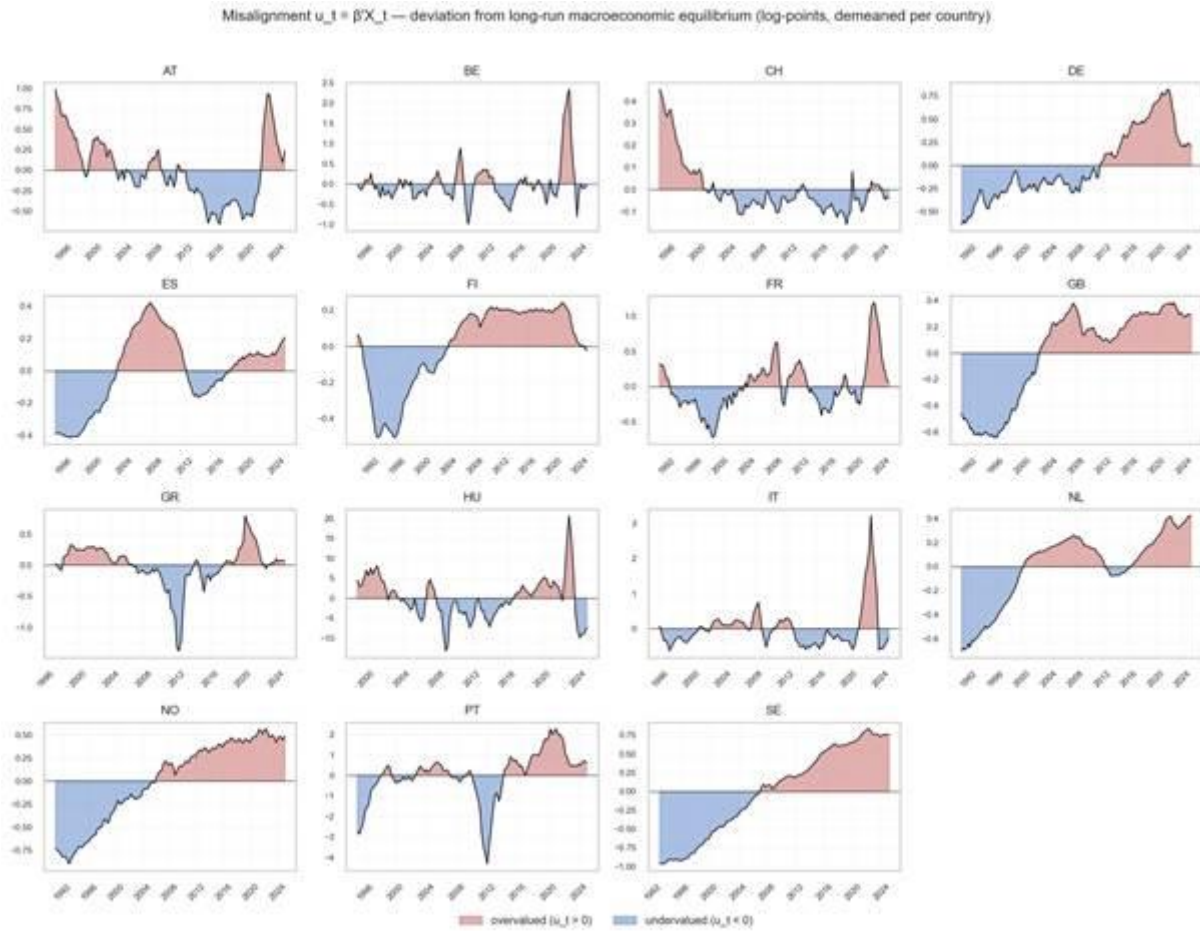
Contributions to Annual Change in Log House Prices (VECM Short-Run, Preferred Specification)



Note. Stacked bars show the decomposition of the annual change in log house prices into contributions from the own-lag (light blue), GDP per capita (dark blue), inflation (orange), 10-year yield (yellow), and the long-run adjustment plus residual (grey). The black line is the realised annual $\Delta \log HP$. Countries with large grey contributions display VECM short-run dynamics that are dominated by the long-run correction term and the residual; countries with prominent macro components (GDP, yield) show explicit fundamental drivers in the short-run dynamics.

Figure H5

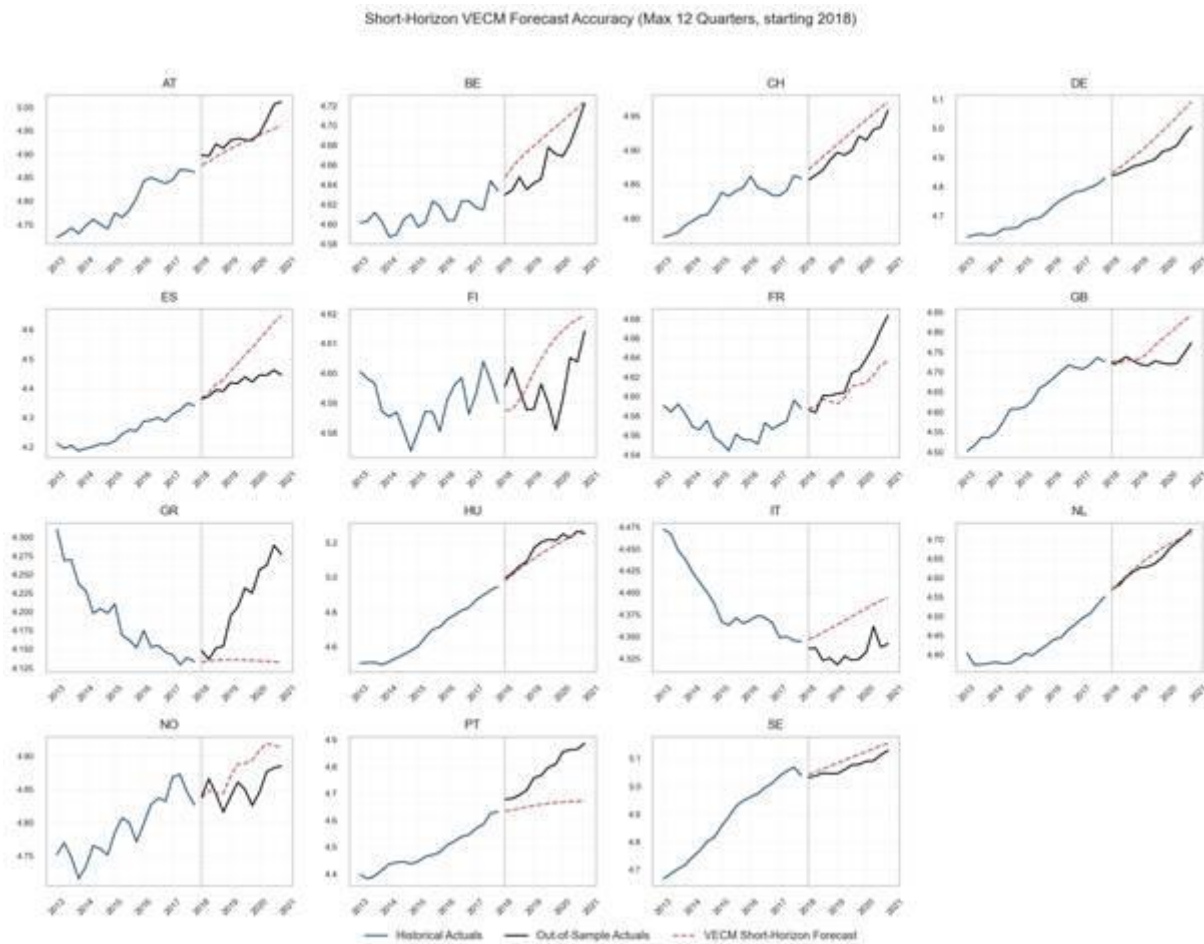
Misalignment Signal $u_t = \beta'X_t - \text{Country Mean (Demeaned per Country)}$



Note. Misalignment series defined as the centred cointegration residual $u_t = \beta'X_t - \text{mean}_{t \in \text{sample}}(\beta'X_t)$. Red shading marks periods of upward deviation from the long-run macro equilibrium (overvaluation in the model's sense), blue shading marks downward deviation (undervaluation). Magnitudes are in log points (a value of 0.20 $\approx$ 22% deviation in level terms, $\exp(0.20) - 1$). The Hungary panel uses the nominal specification with a 26-year sample and exhibits the largest log-point swings; its values must be read qualitatively only (see §5.3 and §6.8).

Figure H6

Short-Horizon VECM Forecast vs. Realised Path per Country (Backtest 2018Q1–2020Q4, log level)



Note. Pseudo-out-of-sample forecast paths over a 12-quarter horizon, holding the cointegration structure fixed at the values estimated up to 2017Q4. Solid blue: training-window historical actuals; solid black: out-of-sample realised log house prices; dashed red: VECM short-horizon forecast. The vertical dotted line marks the train/test cutoff. The plot illustrates the country-by-country path discussion in §6.4 and underpins the RMSE/MAE metrics in Figure H7.

Figure H7

Backtest Accuracy — Actual vs. Forecasted Cumulative Growth (Short-Horizon VECM, 2018–2020)

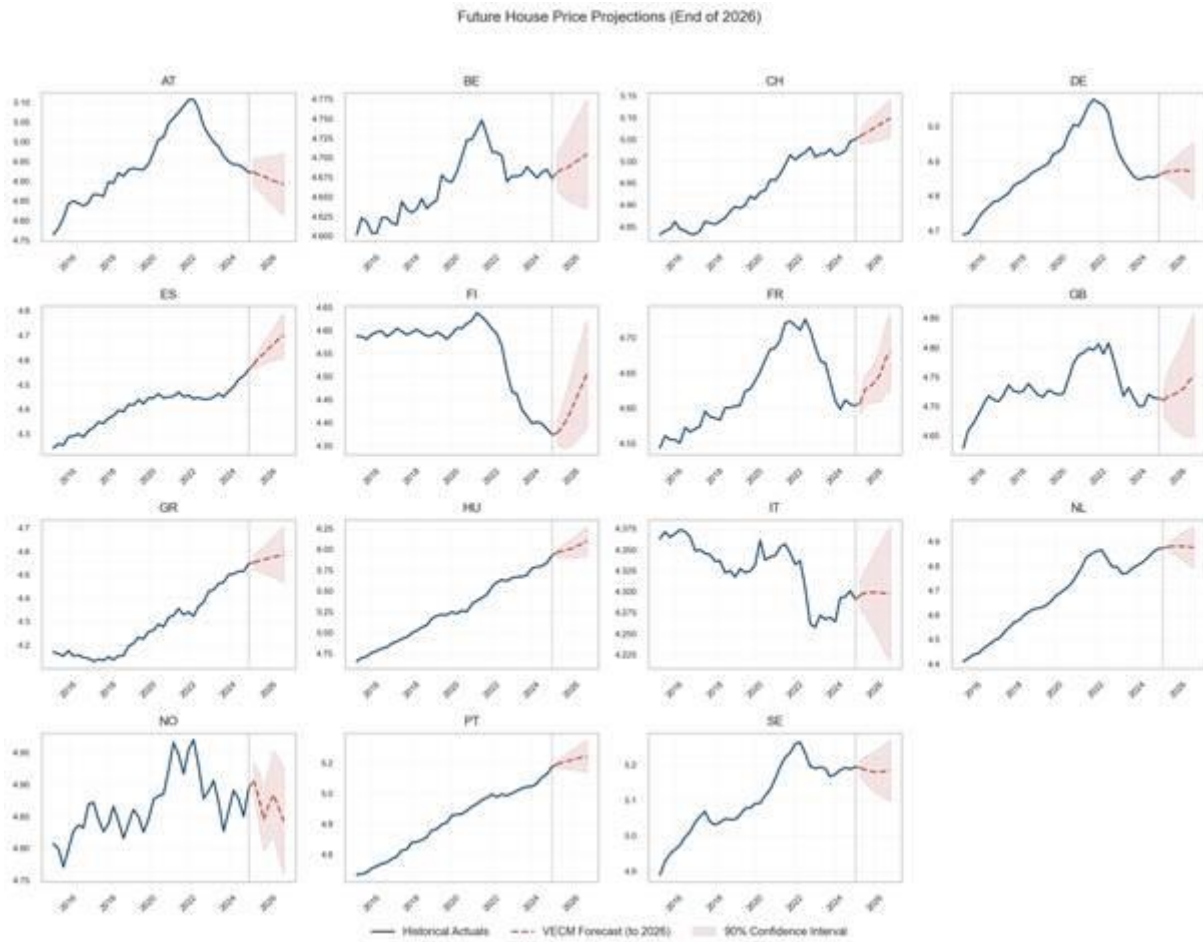
Backtest Accuracy — Actual vs. Forecasted Growth (Short-Horizon VECM, 2018–2020)

Country	Period	Actual Growth %	Forecast Growth %	Forecast Miss %	RMSE	MAE
AT	2018-01-01 - 2020-10-01	12.0	6.4	-5.6	0.0257	0.0201
BE	2018-01-01 - 2020-10-01	9.7	9.9	0.2	0.0275	0.025
CH	2018-01-01 - 2020-10-01	10.7	12.1	1.4	0.0209	0.0199
DE	2018-01-01 - 2020-10-01	18.3	28.9	10.6	0.0646	0.0575
ES	2018-01-01 - 2020-10-01	8.6	32.9	24.3	0.105	0.0841
FI	2018-01-01 - 2020-10-01	1.9	2.4	0.5	0.0151	0.0131
FR	2018-01-01 - 2020-10-01	10.2	5.2	-5.0	0.0215	0.0165
GB	2018-01-01 - 2020-10-01	5.1	12.5	7.4	0.0519	0.0406
GR	2018-01-01 - 2020-10-01	13.8	-1.5	-15.3	0.0928	0.0767
HU	2018-01-01 - 2020-10-01	29.2	32.4	3.1	0.0357	0.0287
IT	2018-01-01 - 2020-10-01	0.5	6.0	5.4	0.0409	0.0382
NL	2018-01-01 - 2020-10-01	16.8	15.5	-1.3	0.0141	0.0111
NO	2018-01-01 - 2020-10-01	4.8	7.7	2.9	0.0372	0.0316
PT	2018-01-01 - 2020-10-01	23.2	-0.6	-23.8	0.1373	0.1228
SE	2018-01-01 - 2020-10-01	10.4	12.9	2.5	0.0288	0.0271

Note. Tabulated rendering of the short-horizon backtest with country-level actual versus forecasted cumulative growth (in percent), forecast miss (forecast minus actual, in percentage points), RMSE and MAE on log levels. The "Forecast Miss %" column is heat-coded: green for over-prediction, red for under-prediction; magnitude scales with colour saturation. This tabular view complements the visual paths in Figure H6.

Figure H8

Future House Price Projections to End-2026 (Conditional VECM Path, 90% CI)



Note. Forward-looking VECM projection of the log house-price level per country, estimated on the full available data sample. Solid blue: historical actuals (last ten years shown); dashed red: conditional point forecast to 2026Q4; red shading: 90% pointwise confidence band. The vertical dotted line marks the start of the projection window. Country-specific spec choice (Real or Nominal) follows §5.3 and is reported alongside Figure H9.

Figure H9

Future House Price Projections to End-2026 — Cumulative Expected Growth by Country (VECM, 90% CI)

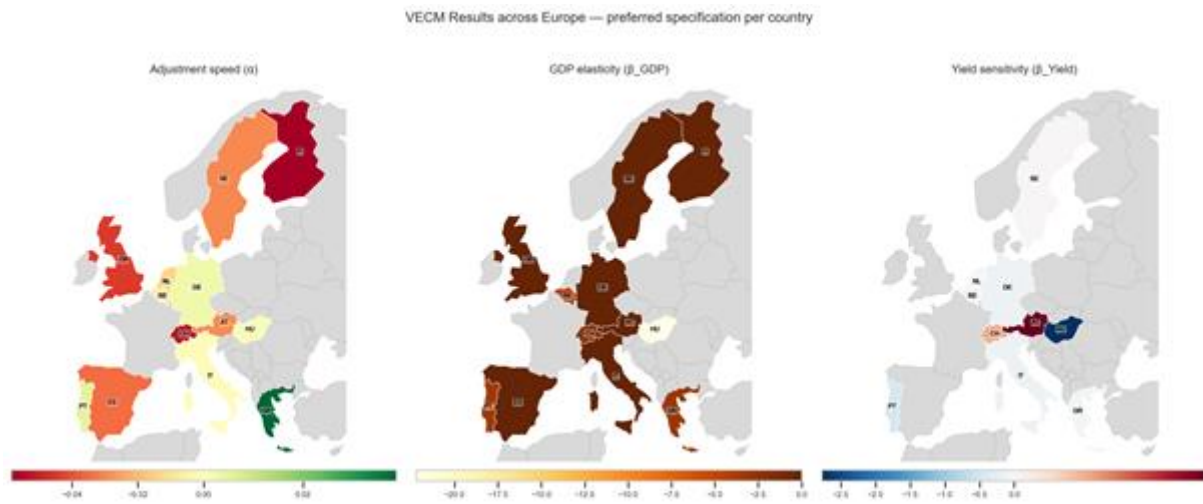
Future House Price Projections to 2026 — ranked by expected growth (VECM, 90 % CI)

Country	Spec	Latest Data	Forecast End	Expected Growth %	Pessimistic (90 CI)%	Optimistic (90 CI)%
FI	Real	2025-07-01	2026-10-01	15.5	2.8	29.8
HU	Nominal	2025-07-01	2026-10-01	11.3	-6.8	32.9
ES	Real	2025-07-01	2026-10-01	8.7	-0.8	19.1
FR	Real	2025-07-01	2026-10-01	7.3	1.6	13.4
GB	Real	2025-10-01	2026-10-01	5.0	-5.6	16.9
GR	Real	2025-07-01	2026-10-01	1.9	-9.7	15.0
CH	Real	2025-10-01	2026-10-01	1.5	-2.8	5.9
BE	Real	2025-07-01	2026-10-01	0.7	-6.1	7.9
DE	Real	2025-07-01	2026-10-01	0.6	-7.6	9.4
PT	Real	2025-07-01	2026-10-01	0.3	-9.8	11.6
NL	Real	2025-07-01	2026-10-01	-0.5	-8.7	8.4
SE	Nominal	2025-10-01	2026-10-01	-1.5	-9.5	7.4
IT	Real	2025-07-01	2026-10-01	-1.8	-9.4	6.4
AT	Real	2025-07-01	2026-10-01	-2.6	-10.0	5.5
NO	Real	2025-10-01	2026-10-01	-3.4	-11.2	5.1

Note. Tabulated rendering of the forward-looking VECM projection in Figure H8, ranked by point-estimate cumulative growth from the latest available data to forecast end. Columns report the country, preferred specification (Real or Nominal per §5.3), latest observed quarter, forecast end date, point-estimate expected growth (%), and the lower (Pessimistic) and upper (Optimistic) bounds of the 90% pointwise confidence band, both expressed as cumulative growth (%). The "Expected Growth %" column is heat-coded: green for projected appreciation, red for projected depreciation; magnitude scales with colour saturation. The wide CI ranges — particularly for Hungary and countries with shorter usable samples — reflect cumulative innovation variance over the projection horizon and should be read alongside the country-by-country path in Figure H8.

Figure H10

Cross-Country VECM Estimates across Europe (Preferred Specification per Country)



Note. Visual rendering of the three central VECM coefficients for the preferred per-country specification. Left panel: speed of adjustment α on the house-price equation, with diverging colour scale centred at zero — green shading marks countries where prices themselves move toward the long-run equilibrium following a deviation, red shading marks countries where adjustment occurs primarily through other system variables. Middle panel: long-run GDP elasticity β_{GDP} from the first cointegration vector. Right panel: yield sensitivity β_{Yield} . Grey-shaded countries either lie outside the panel or are excluded by the $I(1)$ filter (Denmark). Norway is shown in grey on the β_{Yield} panel where the coefficient is normalised to zero by construction.

Author contributions

Author	Contribution
Martin	Joint contribution across the full paper: research question, hypotheses, country selection and structure; BIS, FRED and World Bank data collection, harmonisation, checking and quarterly preparation; ADF tests, Johansen tests, VECM estimates, lag selection and robustness checks; interpretation of country results, Solid and Weak classifications, the misalignment signal and investment relevance; writing, revision and final alignment of the introduction, literature, methodology, results and conclusion; tables, figures, appendix and reproducible documentation.
Marc	Joint contribution across the full paper: research question, hypotheses, country selection and structure; BIS, FRED and World Bank data collection, harmonisation, checking and quarterly preparation; ADF tests, Johansen tests, VECM estimates, lag selection and robustness checks; interpretation of country results, Solid and Weak classifications, the misalignment signal and investment relevance; writing, revision and final alignment of the introduction, literature, methodology, results and conclusion; tables, figures, appendix and reproducible documentation.
Mark	Joint contribution across the full paper: research question, hypotheses, country selection and structure; BIS, FRED and World Bank data collection, harmonisation, checking and quarterly preparation; ADF tests, Johansen tests, VECM estimates, lag selection and robustness checks; interpretation of country results, Solid and Weak classifications, the misalignment signal and investment relevance; writing, revision and final alignment of the introduction, literature, methodology, results and conclusion; tables, figures, appendix and reproducible documentation.

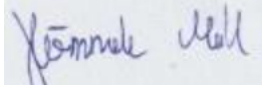
Use of artificial intelligence

Purpose of use	Description of what AI tools were used for	Affected parts of the work	AI tools used
Code	Support with Python scripts for data collection, data cleaning, ADF tests, Johansen tests, VECM estimates, robustness checks and graphical analyses.	Data collection, data cleaning, data analysis, results and appendix	Claude Code
Grammar and sentence structure	Language revision, shortening and simplification of individual sections, plus support with grammar, sentence structure and clear wording.	Introduction, related literature, hypotheses, research design, results and conclusion	ChatGPT
Presentation	Support with document structure, headings, tables, figures and the compact presentation of key results.	Chapter structure	Claude Design
Literature	Support with search terms, thematic narrowing and identification of suitable sources on house prices, misalignment, cointegration, VECM and macroeconomic drivers.	Related literature, theoretical foundation and methodological justification	Gemini, Claude, ResearchGate, Google Scholar

We, Martin Riemann / Marc Bauer / Mark Hämmerle / hereby certify that the attached work, is wholly and completely our own and that we have indicated all the sources (printed, electronic, personal, etc.) that we have consulted. Any sections quoted from these sources are clearly indicated in quotation marks or are otherwise so declared. We further attest that we have included acknowledgement of the name(s) of any person(s) consulted in the course of preparing this assignment.

10/05/2026	Martin Riemann	
Date	Name	Signature

10/05/2026	Marc Bauer	
Date	Name	Signature

10/05/2026	Mark Hämmerle	
Date	Name	Signature